# Title of the Thesis

Optional Subtitle of the Thesis

Author Full Name

Master in

**Programme Name**

Supervisor:
Prof. Supervisor Name, PhD,
Position, Department,
Iscte – Instituto Universitário de Lisboa

Co-Supervisor:
Prof. Co-Supervisor Name, PhD,
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Position, Department,
Institution

Month, Year

*To my family.*

# Acknowledgments

I would like to thank my supervisor, **Prof. Supervisor Name**, for the guidance and support throughout this work.

# Resumo

Este trabalho aborda o problema de …

**Palavras-chave:** palavra-chave 1, palavra-chave 2, palavra-chave 3.

# Abstract

This thesis addresses the problem of …

**Keywords:** keyword 1, keyword 2, keyword 3.

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# List of Acronyms

| Acronym | Definition |
|---|---|
| ACDC | Automated Cardiac Diagnosis Challenge |
| AHA | American Heart Association |
| ASSD | Average Symmetric Surface Distance |
| DSC | Dice Similarity Coefficient |
| DCM | Dilated Cardiomyopathy |
| ED | End-Diastole |
| EDT | Euclidean Distance Transform |
| ES | End-Systole |
| GNN | Graph Neural Network |
| HCM | Hypertrophic Cardiomyopathy |
| HD95 | 95th-percentile Hausdorff Distance |
| ICC | Intraclass Correlation Coefficient |
| INR | Implicit Neural Representation |
| IoU | Intersection over Union |
| LV | Left Ventricle |
| LoA | Limits of Agreement |
| MAE | Mean Absolute Error |
| MRI | Magnetic Resonance Imaging |
| NOR | Normal cohort |
| PCA | Principal Component Analysis |
| PDE | Partial Differential Equation |
| PRISMA | Preferred Reporting Items for Systematic Reviews and Meta-Analyses |
| RMSE | Root Mean Squared Error |
| RV | Right Ventricle |
| SAX | Short-Axis |
| SDF | Signed Distance Function |
| SSM | Statistical Shape Model |

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CHAPTER 1

# Introduction

## 1.1. Motivation

Understanding cardiac structure is fundamentally a geometric problem. The left ventricle (LV), the chamber that pumps oxygenated blood into the body, is a continuously varying three-dimensional structure whose shape, size, and wall thickness change throughout the cardiac cycle. Each heartbeat is a cycle of filling and contraction. At end-diastole (ED) the ventricle has filled with blood and is at its largest; at end-systole (ES) it has contracted and expelled part of that blood, so the cavity is smallest and the muscular wall, the myocardium, is at its thickest. These two instants, or cardiac phases, are the ones routinely segmented and measured in clinical practice, because the change between them quantifies how well the ventricle pumps. The shape of the ventricle and the thickness of its wall at these phases provide important information about cardiac function and about structural remodelling, the changes in shape and wall thickness through which the heart adapts to disease; changes in ventricular geometry and myocardial thickness are associated with several forms of cardiac disease [1, 2, 3].

Cardiac magnetic resonance imaging (MRI) provides detailed observations of this anatomy, but these observations are acquired as a set of two-dimensional slices rather than as a complete three-dimensional representation. In a short-axis (SAX) acquisition, the LV is imaged in parallel planes perpendicular to its long axis, from the base, the wide end where the valves sit, towards the apex, the pointed tip. On each slice the endocardium, the inner boundary of the blood-filled cavity, and the epicardium, the outer boundary of the myocardium, can be delineated; this outlining is called segmentation. Within each slice, the spatial resolution can be relatively high, while the distance between adjacent slices is substantially larger. Consequently, a structure that is anatomically continuous is represented by a limited number of discrete observations.

This distinction becomes important when measurements are extended beyond the acquired slices. Myocardial wall thickness, for example, is not simply a property of isolated contours. It varies across the ventricle (local thickness) and can reflect changes in cardiac structure, including myocardial hypertrophy, an abnormal thickening of the wall [1]. Estimating such quantities from sparse observations therefore depends not only on the accuracy of the segmentation, but also on how well the underlying three-dimensional geometry is represented.

A continuous reconstruction can provide a more coherent description of the LV and enable measurements in regions that are not directly observed. Such representations are particularly relevant in large imaging studies, where consistent anatomical descriptions are needed for automated analysis and comparison across subjects [4, 5]. They can also support regional analysis of myocardial structure, including measurements organised according to the American Heart Association 17-segment model [6].

The central challenge addressed in this thesis is therefore not the extraction of LV contours alone, but the inference of the three-dimensional geometry that lies between sparse observations. A useful reconstruction must recover continuous endocardial and epicardial surfaces while preserving patient-specific shape and maintaining a anatomically plausible myocardial wall. Simple interpolation or surface lofting can provide continuity, but does not necessarily recover the geometry that best represents the underlying anatomy.

This thesis investigates whether a phase-conditioned implicit neural representation can learn this reconstruction problem from synthetic LV geometries and meshes derived from real cardiac MRI segmentations. An implicit neural representation is a neural network that describes a surface indirectly, as the set of points where a function it computes is zero; the one proposed here is also told which cardiac phase it is reconstructing. The proposed approach thus represents the LV surfaces as continuous functions in three-dimensional space, allowing geometry to be queried at arbitrary locations rather than only at the acquired slices. The reconstructed surfaces are subsequently used to study myocardial wall thickness and to compare the resulting measurements with those obtained from surfaces fitted to the same clinical contours with a conventional smooth-surface method based on radial basis functions (RBFs). The aim is not to replace clinical interpretation, but to investigate whether continuous geometric representations can provide a more consistent basis for quantitative analysis of sparse cardiac MRI. Figure 1.1 summarises this pipeline, from the segmented SAX slices of both phases to the reconstructed surfaces, their local wall-thickness maps, and a regional summary of systolic wall thickening.

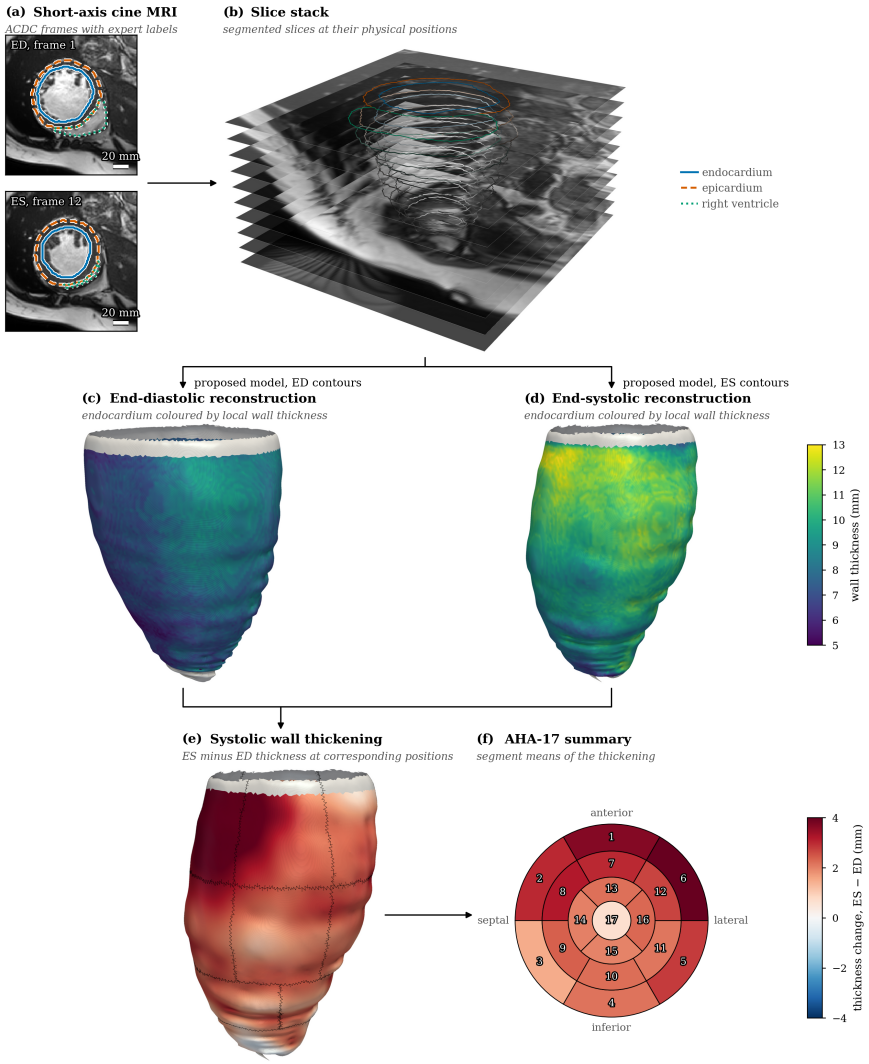



**Figure 1.1.** Overview of the proposed pipeline. (a) SAX cardiac MRI slices at end-diastole (ED) and end-systole (ES), segmented into endocardium, epicardium, and right ventricle. (b) The segmented slices placed at their physical positions along the long axis of the LV, forming a sparse stack. The LV contours of each phase are given to the proposed model, which reconstructs continuous endocardial and epicardial surfaces for (c) ED and (d) ES; the endocardium is coloured by the local myocardial wall thickness. (e) The two phases are combined to show how much the wall thickens from ED to ES at each position on the surface, with the AHA 17-segment boundaries drawn in black. (f) The same thickening summarised per segment in the AHA 17-segment bullseye.

## 1.2. Problem Statement

SAX cardiac MRI provides detailed cross-sectional information about the LV, but does not directly provide a complete three-dimensional representation. The slices are separated along the ventricular long axis, leaving portions of the geometry unobserved between adjacent acquisitions. As a result, the spatial resolution within each slice is typically much finer than the resolution between slices. Reconstructing a continuous surface from these sparse observations is therefore an ill-posed geometric problem: many different surfaces are consistent with the same set of slices.

A simple connection of the segmented contours may produce a continuous surface, but continuity alone does not guarantee a plausible reconstruction. Such approaches can introduce irregularities,

gaps, or local distortions, especially when the distance between slices is large. The challenge is particularly important when both the endocardial and epicardial surfaces are considered, because they must be reconstructed as a consistent pair. The endocardium defines the inner boundary of the LV cavity, while the epicardium defines the outer boundary of the myocardium. Their relative position must remain physiologically plausible so that the reconstructed wall retains positive and spatially coherent thickness.

The problem is further constrained by the characteristics of publicly available cardiac MRI datasets. These datasets commonly provide images and segmentation masks for individual slices, label images in which each pixel is marked as cavity, myocardium, or background, rather than complete three-dimensional LV meshes, that is, surfaces made of connected triangles. Meshes are nevertheless useful for directly training and evaluating surface-reconstruction models because they provide an explicit representation of the continuous geometry. A method is therefore required to bridge the gap between sparse slice-based observations and continuous three-dimensional surfaces.

This thesis addresses this problem using a phase-conditioned implicit neural representation (INR) trained on synthetic LV shapes and meshes derived from real cardiac MRI segmentations. The model receives sparse contours as input and represents the endocardial and epicardial surfaces as continuous three-dimensional functions. Conditioning the representation on cardiac phase, that is, telling the model whether the contours come from an ED or an ES frame, allows it to distinguish between end-diastolic and end-systolic geometry, which differ markedly in cavity size and wall thickness. In addition, the reconstruction is constrained so that the outer surface remains outside the inner surface, preserving a positive myocardial wall thickness.

The reconstructed geometry is evaluated at both the surface and measurement levels. Geometric agreement is assessed against reference surfaces, while myocardial wall thickness is measured on the reconstructed geometry and compared with measurements obtained from the RBF-fitted clinical surfaces introduced above. This makes it possible to evaluate not only whether the proposed model produces visually continuous surfaces, but also whether the resulting geometry preserves quantities relevant to cardiac analysis.

## 1.3. Research Questions

This thesis addresses the following research questions:

- **RQ1.** To what extent can a continuous, watertight three-dimensional surface of an anatomical structure be reconstructed from sparse two-dimensional contours, and how does this reconstruction compare with conventional implicit and shape-model approaches applied to the same contours?
- **RQ2.** Can a single model represent multiple physiological states of the same structure while preserving a physically plausible separation between its inner and outer surfaces?
- **RQ3.** To what extent do measurements derived from the reconstructed geometry agree with those obtained from reference clinical geometry, and how does that agreement depend on the measurement method?

CHAPTER 2

# Literature Review

This chapter describes the review methodology and examines prior work on reconstructing paired LV surfaces from sparse cardiac MRI observations and measuring the myocardial wall represented by those surfaces.

## 2.1. Review Methodology

The review was designed to compare methods relevant to sparse cardiac MRI, three-dimensional shape reconstruction, geometric learning, and myocardial wall-thickness measurement, rather than to provide an exhaustive systematic review.

Scopus, PubMed, and IEEE Xplore were searched, with Google Scholar used for backward and forward citation chaining. The search included foundational work published from 2000 onwards and selected studies available up to 2026. Although recent work was prioritised, earlier publications were retained where they establish concepts used by later methods. The resulting review is selective and does not claim exhaustive coverage of the literature published through 2026.

Search terms combined anatomy and imaging concepts (“cardiac MRI”, “left ventricle”, “short-axis”, “SAX”, “myocardium”) with reconstruction terms (“3D reconstruction”, “mesh reconstruction”, “shape model”, “implicit surface”, “signed distance function”), learning terms (“deep learning”, “graph neural network”, “point cloud”, “neural representation”), and measurement terms (“wall thickness”, “myocardial thickness”, “Laplace equation”, “transmural”).

The refined query applied consistently across databases was:
*(“cardiac MRI” OR “left ventricle” OR “short-axis”) AND (“3D reconstruction” OR “statistical shape model” OR “signed distance function” OR “graph neural network” OR “implicit representation”) AND (“wall thickness” OR “mesh” OR “deep learning”)*

Studies were included when they addressed LV three-dimensional reconstruction or shape modelling, proposed or evaluated myocardial wall-thickness methods, developed geometric learning methods applicable to anatomical meshes, or introduced benchmark cardiac MRI data relevant to LV analysis. Studies focused only on segmentation, on non-cardiac anatomy without transferable methodology, or lacking peer review or sufficient methodological detail were excluded. Publications before 2000 were considered only when they provided foundational methods directly relevant to the reviewed approaches.

The search returned 142 records. After duplicate removal and screening, 28 studies met the selection criteria; backward citation searching added 7, giving a final corpus of 35 publications. Figure 2.1 summarises this process.

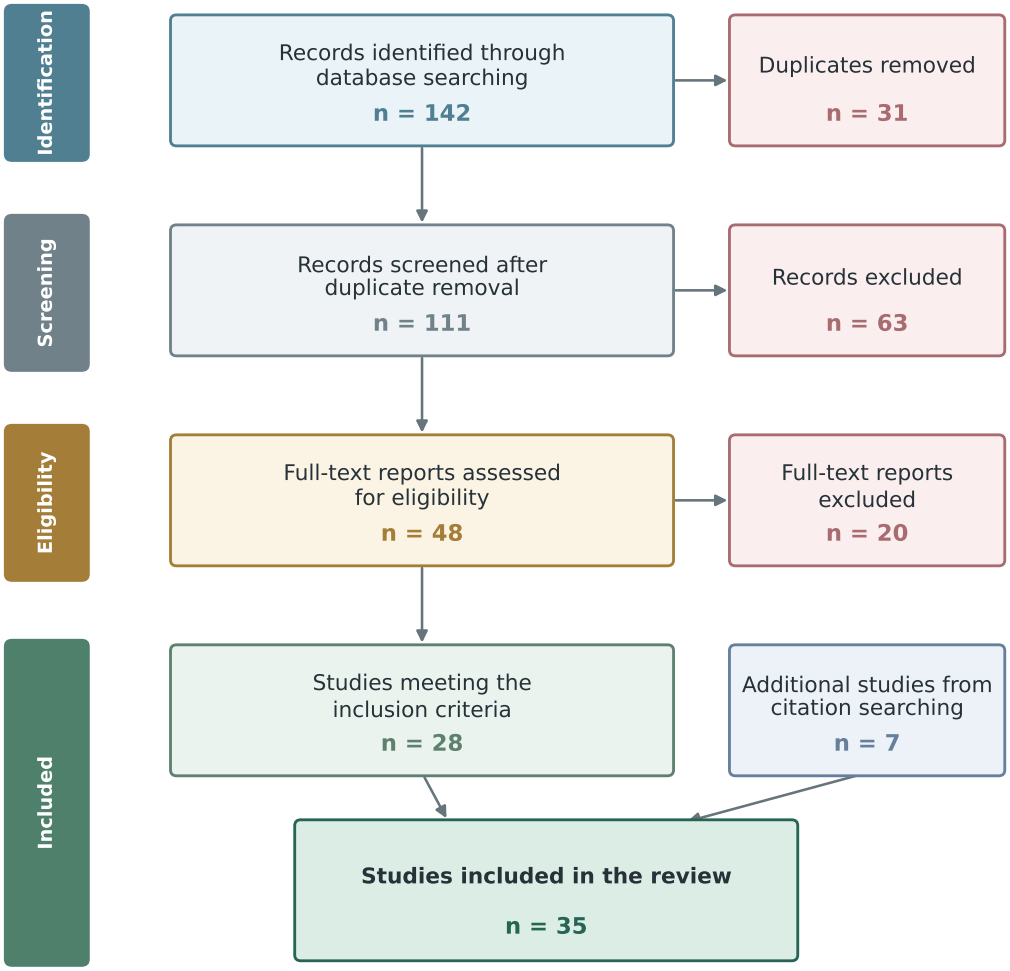



**Figure 2.1.** PRISMA flow diagram summarising the selection of the 35 publications included in the review.

The selected studies cover cardiac MRI data and segmentation, statistical shape models, learned 3D reconstruction, and wall-thickness measurement. The next section relates these areas to the specific problem of recovering paired LV surfaces from sparse contours.

## 2.2. Technical Background

The central distinction in this review is between anatomy observed on acquired MRI slices and geometry that must be inferred between them. This distinction connects the constraints of paired LV surface reconstruction to the available representations of three-dimensional anatomy.

Cine cardiac MRI captures the LV at several points in the cardiac cycle. Of these, end-diastole (ED), when the cavity is largest, and end-systole (ES), when it is smallest, provide a useful description of ventricular contraction and are the phases labelled in benchmarks such as ACDC [7].

In a SAX acquisition, each plane samples a cross-section perpendicular to the ventricular long axis. The anatomy is resolved in detail within a slice, but the larger separation between consecutive slices produces an anisotropic image stack [8]. Segmentation can therefore recover the myocardium where it is observed without determining how its boundaries curve, taper, or change in the unobserved intervals.

Each slice contributes an endocardial contour around the cavity and an epicardial contour around the outer myocardium. In three dimensions, these must form nested surfaces with a positive separation. A reconstruction that treats the surfaces independently may produce individually plausible boundaries while still creating an intersection, reversal, or local collapse in the wall between them, a difficulty also addressed by recent joint implicit reconstructions of cardiac structures [9].

Recovering the continuous geometry is consequently underdetermined: many smooth surfaces can pass through the same contour rings. Smoothing stabilises the reconstruction but can remove genuine regional variation; a strong population prior can bridge missing regions but may pull abnormal anatomy towards common shape; and close contour fitting can preserve segmentation noise. A useful method must balance these competing constraints rather than optimise any one of them in isolation [8, 10].

The same reasoning determines how reconstruction should be evaluated. The surfaces should agree with observed anatomy, remain closed where volume is needed, preserve subject-specific shape, and form a valid wall at both phases. No single overlap or surface-distance measure assesses all of these properties, so geometric distance, volumetric agreement, topology, and the stability of wall-derived measurements must be considered together [9, 10].

The representation chosen for three-dimensional anatomy determines both what can be measured and how missing geometry is inferred. Voxel volumes are well suited to image-based segmentation and overlap analysis, but retain the spatial limits of the sampling grid. Surface meshes support direct geometric and regional measurements, although remeshing, smoothing, and repair can alter the anatomy being evaluated. Neither representation alone determines the geometry between sparsely acquired slices.

Statistical shape models address this missing information by constraining meshes with population-level modes of variation [10, 11]. They can therefore provide a geometric prior or synthetic training examples, but remain limited by the topology and anatomical variation represented in the source atlas.

Implicit methods offer a different compromise. DeepSDF and Occupancy Networks represent geometry as continuous implicit fields [12, 13], from which meshes can be extracted using methods such as Marching Cubes [14]. Work on Eikonal regularisation and Fourier features further addresses distance-field consistency and the recovery of local spatial variation [15, 16]. By improving field consistency and spatial detail, these developments make implicit representations relevant to sparse-contour reconstruction. The extracted surface nevertheless remains dependent on the learned field and extraction procedure.

Paired LV surfaces introduce an additional constraint: two independently predicted fields do not guarantee a valid ordering. Defining one surface in relation to the other through a positive separation instead represents a reference boundary and a constrained wall. Joint implicit reconstruction of cardiac structures provides one way of modelling related boundaries together, although surface ordering and wall validity must still be assessed explicitly [9].

## 2.3. Related Work

This section compares how the reviewed methods address the evidence available on acquired slices, the geometry missing between them, and the measurements derived from the reconstructed wall.

### 2.3.1. Clinical observations and conventional geometric priors

Short-axis (SAX) cine MRI is the standard protocol for left-ventricle (LV) evaluation, producing 2D slices perpendicular to the long axis with high in-plane resolution (1–2 mm) but large inter-slice gaps (6–10 mm) [8]. The resulting anisotropy creates the central reconstruction problem: the observed contours are detailed within each slice, but the geometry between slices, especially near the apex and base, must be inferred. The AHA 17-segment model provides a common anatomical frame for reporting regional measurements [6].

Modern segmentation networks can identify endocardial and epicardial boundaries reliably on the acquired slices. U-Net established the encoder–decoder pattern used by many medical segmentation systems, while nnU-Net showed that much of their configuration can be adapted automatically to a dataset [17, 18].

Although both methods improve the observed contours, they still optimise agreement on sampled image planes. High slice-wise overlap can therefore coexist with uncertain through-plane geometry. Moreover, because the contours become constraints on the reconstructed surface, segmentation errors can propagate into regions where no image evidence is available rather than remaining local to one slice.

ACDC provides expert ED and ES labels across five diagnostic groups, including normal and hypertrophic ventricles [7]. Its phase and diagnostic variation make it useful for testing whether a reconstruction retains contraction-related and abnormal shape differences. The M&Ms-2 challenge focuses on right-ventricular segmentation and extends evaluation across centres, scanner vendors, and diseases [19]. For reconstruction, its multi-source design shows that performance on one acquisition source does not guarantee transfer to another. A plausible surface on one cohort is insufficient evidence of generalisation because acquisition variation affects the contours from which the entire surface is inferred.

At population scale, the UK Biobank provides a large and standardised cardiac MRI resource [20]. Automated segmentation of this cohort has enabled large anatomical studies and statistical atlases [5]. Together, slice-wise clinical observations and population-level atlases provide complementary forms of anatomical evidence.

The two forms are not interchangeable. Clinical labels retain subject-specific and pathological variation, but their through-plane geometry is only indirectly observed. A population atlas supplies complete and consistent surfaces, but it reflects the cohort and processing choices from which it was built. Combining them can reduce the weakness of either source alone, provided that the atlas is used as a prior rather than treated as patient-specific ground truth.

Statistical Shape Models (SSMs) encode anatomical variability by aligning a training population to point-to-point correspondence and applying PCA to extract the dominant modes of variation [10, 11]. They provide a compact description of plausible shape, but their quality depends on anatomical correspondence, alignment, and the diversity of the population used to construct them.

Cardiac SSMs have progressed from multi-object ventricular models to large population atlases. Dense correspondence allows coupled structures to be represented consistently across subjects [21], while the UK Digital Heart Project atlas captures population variation in ventricular shape and motion [4].

Their modelling of coordinated variation across related cardiac structures is directly relevant to paired-surface reconstruction, where the two boundaries cannot be treated in isolation.

Shape-model studies have also identified variation associated with hypertrophic remodelling, normal anatomical diversity, and myocardial infarction [2, 3, 22]. Their findings show that clinically relevant anatomy is not captured by global volume alone. A population prior must therefore retain regional differences in proportion, apical form, and curvature rather than draw every case towards one mean geometry.

SSMs have two principal roles. They can constrain reconstruction to a learned shape space, or they can generate synthetic geometries by sampling that space [10, 23]. Synthetic sampling provides complete, plausible meshes that can increase the amount of geometric supervision available for learning-based methods.

Synthetic sampling increases the amount of complete geometric supervision, but it cannot introduce anatomy absent from the source atlas. It may also under-represent rare or severe pathology when the retained PCA modes favour common variation. Clinical data remain important for adapting a population prior to subject-specific and pathological anatomy while retaining the surface continuity learned from complete meshes. Combining SSM-derived and clinical data therefore assigns complementary roles to population-level priors and observed anatomical variation.

Before a learned model can be supervised with clinical data, the segmented SAX contours must be related to a three-dimensional surface. Linear contour interpolation, SSM fitting, and Radial Basis Function (RBF) fitting offer three different ways to supply the missing through-plane geometry. Their principal difference is the source of the reconstruction prior: linear interpolation uses only neighbouring contours, an SSM uses population-level modes of anatomical variation, and an RBF fit imposes smoothness through a continuous implicit field.

Linear contour lofting first establishes correspondence between points on adjacent rings and then joins those points with triangular strips. It is transparent, computationally inexpensive, and passes directly through the observed contours. However, the geometry between two slices is piecewise linear and depends on the chosen starting-point alignment and point correspondence. Large inter-slice gaps therefore produce abrupt changes in slope, while the apex and base require explicit closure rules. These limitations make lofting useful as a direct interpolation baseline, but less suitable as the sole source of smooth three-dimensional supervision from anisotropic SAX data [8].

SSM fitting instead aligns a template with the observations and optimises its shape coefficients so that the deformed surface approaches the contours. The result retains a fixed topology and population-informed anatomical plausibility, which are valuable when observations are sparse [10, 11]. The same prior also limits the fit: anatomy outside the variation represented by the atlas cannot be recovered faithfully, and fitting requires a model appropriate to the structure and cardiac phase under study. An ED-only SSM cannot provide equivalent supervision for ES without imposing a diastolic shape space on systolic observations.

RBF reconstruction represents the surface as the zero level set of a smooth function fitted to scattered points [24]. Surface samples are assigned zero values, while additional samples displaced to either

side of the contour provide the sign needed to distinguish interior from exterior. The resulting implicit field can be evaluated on a regular grid. Marching Cubes then extracts a mesh from the sampled field [14].

Unlike SSM fitting, this procedure does not require a phase-specific template or fixed mesh correspondence. Unlike linear lofting, it defines the intervening geometry as a smooth field rather than as strips between paired vertices. Its limitations are different: the reconstruction depends on the kernel, smoothing, and off-surface constraints, and strong regularisation can suppress genuine local variation.

### 2.3.2. Deep learning models for 3D shape reconstruction

Classical reconstruction methods depend on the sampled contours and on the assumptions imposed between slices, particularly near the apex and base, as discussed in Section 2.3.1. Learned reconstruction methods address the same trade-off between observation fidelity and geometric prior through two main families: template deformation and implicit field prediction.

Template methods learn to displace the vertices of a fixed mesh. Pixel2Mesh and Pose2Mesh demonstrate that graph-based deformation can recover dense geometry from sparse image or landmark evidence [25, 26]. In cardiac applications, fixed correspondence is valuable for population analysis and regional measurement [4, 27, 28]. The trade-off is that the output remains tied to the topology and resolution of the chosen template. The same constraint that simplifies correspondence can therefore limit how far the reconstruction departs from the anatomy encoded by that template.

Implicit Neural Representations (INRs) remove the fixed output template by learning a continuous field from which a surface can be extracted at a chosen resolution. DeepSDF and Occupancy Networks established this approach using signed-distance and occupancy formulations, respectively [12, 13]. For sparse cardiac reconstruction, removing the predefined mesh avoids restricting the output geometry to fixed resolution and connectivity. Output flexibility does not, however, guarantee that two predicted LV surfaces remain correctly ordered; anatomical validity must be addressed separately.

Recent cardiac work shows why implicit representations are useful when MRI slices are far apart. In a SAX scan, the detailed shape of the ventricle is observed within each slice, but less information is available between slices. Brás et al. [9] addressed this anisotropy problem with a neural implicit representation that reconstructs the left and right ventricles, including the endocardial and epicardial boundaries of the LV myocardium. The study is closely related to sparse-contour reconstruction because both approaches seek continuous cardiac shape from incomplete observations. Brás et al. use image data directly, whereas contour-based approaches operate on segmented point observations.

Muffoletto et al. [29] also reconstruct three-dimensional cardiac shape from sparse segmentations with a neural implicit representation. Their method uses a learned biventricular shape prior and includes both SAX and long-axis views. It is therefore broader than LV-focused SDF reconstruction. At the time of writing, the authors provide a public project repository, although a complete code and dataset release could not be verified.

The INR literature also provides mechanisms relevant to thin cardiac anatomy. Fourier features improve the recovery of local spatial variation, while Eikonal regularisation promotes a consistent distance

field [15, 16]. In sparse-slice settings, both mechanisms can help mitigate the loss of local surface detail associated with interpolation across large gaps.

Recent cardiac INRs also model motion across the cardiac and respiratory cycles [30, 31, 32, 33]. Their main goal is temporally coherent image or volume reconstruction, whereas other implicit approaches focus specifically on continuous anatomical surface reconstruction conditioned on cardiac phase.

PointNet provides a complementary way to encode unordered observations [34]. Its permutation-invariant formulation is relevant to SAX contours because arbitrary point ordering should not affect the inferred anatomy, making it suitable for point-based encoders in implicit surface models.

### 2.3.3. Wall-thickness measurement techniques

Myocardial wall thickness is a primary clinical biomarker for cardiac remodelling. Grossman et al. [1] demonstrated that wall thickness is a principal determinant of left-ventricular diastolic stiffness, with hypertrophied ventricles exhibiting markedly elevated stiffness proportional to end-diastolic wall thickness. More recently, Huelnhagen et al. [35] showed at 7.0T MRI that wall thickness correlates with tissue transverse relaxation properties, confirming its value as a non-invasive surrogate for myocardial tissue characterisation. The AHA 17-segment model [6] provides the standard framework for reporting regional thickness values.

The accuracy of wall-thickness measurement depends critically on the quality of the underlying surface reconstruction, as any smoothing or geometric artefact propagates directly into the thickness estimate. The literature describes five families of measurement methods, each with distinct trade-offs.

The simplest approach uses nearest-neighbour distance: for each endocardial vertex, a KD-tree identifies the closest epicardial point, yielding a measurement in $O(N \log M)$ time. Its runtime depends on mesh size and the implementation. Nearest-neighbour distance can provide a lower bound on the true transmural thickness because it ignores the anatomical direction of measurement — the nearest point may lie tangentially rather than across the wall.

Ray-based methods address this limitation by casting a ray from each endocardial vertex along its surface normal until it intersects the epicardium. The intersection is computed via the Möller–Trumbore algorithm, accelerated with a Bounding Volume Hierarchy (BVH) to $O(\log F)$ per query. Measuring along the surface normal better approximates a transmural direction but remains sensitive to normal noise: a slight rotation can produce a spurious long or short ray. The Shape Diameter Function variant improves robustness by casting a cone of $K$ rays (typically $K = 7$) at a half-angle of $30^\circ$ around the normal and reporting the median intersection distance, effectively filtering outlier hits caused by local mesh irregularities.

Volumetric methods operate on voxelised representations. The Euclidean Distance Transform (EDT) computes, for each myocardial voxel, the distance to the nearest boundary, and the sum of the endocardial and epicardial distance fields approximates local thickness ($t = D_{\text{endo}} + D_{\text{epi}}$). The estimate is exact for planar parallel boundaries but can be biased in curved regions because the nearest-boundary

directions do not necessarily follow a common transmural path. The method is computationally efficient ($O(V)$ via the Saito–Toriwaki algorithm) but limited by voxel resolution.

Partial differential equation (PDE)-based methods are widely used for transmural thickness estimation. In a method developed for cortical rather than cardiac anatomy, Jones et al. [36] solved Laplace's equation between the two tissue boundaries and used the resulting non-intersecting field lines to establish corresponding paths. The same geometric principle can be transferred to myocardial boundaries, with thickness obtained from the path length between the boundaries along these streamlines. Yezzi and Prince [37] formulated an Eulerian PDE method that computes distances propagated from both boundaries along the Laplacian field. Their sum gives the local tissue thickness.

Finally, mesh-based regularised correspondence methods combine an initial geometric estimate (normal projection or nearest neighbour) with iterative Laplacian smoothing on the mesh connectivity graph. At each iteration, each vertex's thickness is updated as a weighted average of its neighbours, anchored to the initial estimate. Regularisation produces spatially smooth thickness maps suitable for bullseye visualisation, at the cost of potentially attenuating real local variation when the smoothing parameter is too aggressive.

Wall-thickness values depend on both the measurement method and the reconstructed surfaces. Smoothing may make a measurement more stable, but it can also change the distance between the endocardial and epicardial surfaces. Similarly, an irregular surface can create thickness variation that does not reflect the myocardium. For this reason, thickness methods should be compared on the same surfaces. Reconstructions should also be compared using the same thickness method. The present work evaluates both reconstruction consistency and agreement between thickness methods.

### 2.3.4. Summary and research gap

The reviewed literature shows that each component of the LV analysis pipeline is well-studied in isolation: deep learning gives accurate slice-wise segmentation, SSMs encode population-level shape priors, implicit representations provide continuous surface modelling, and multiple wall-thickness algorithms exist. However, limited work integrates these components into a single pipeline for sparse SAX contour-based reconstruction.

A gap remains in methods that jointly address sparse SAX contour encoding, continuous paired endocardial and epicardial surface reconstruction, cardiac-phase conditioning, preservation of positive wall separation, and training with both clinical observations and SSM-derived synthetic geometries. Such an integrated framework would combine the complementary strengths of population-based shape priors, continuous implicit representations, and clinical contour data while supporting wall-thickness measurement beyond the acquired slices.

There is also an evaluation gap. When no complete three-dimensional reference is available, contour agreement only tests the acquired slices. It cannot validate the reconstructed shape between slices. Evaluation should therefore distinguish contour fidelity from agreement between reconstruction methods. Thickness values must also be interpreted as measurements on an inferred wall, not as direct observations. These principles guide the evaluation in the following chapters.

CHAPTER 3

# Methodology

This chapter presents the methodology for reconstructing three-dimensional LV geometry from sparse SAX contours and measuring myocardial wall thickness on the resulting surfaces. The approach combines synthetic geometric supervision with clinical MRI-derived data, a phase-conditioned implicit representation, and a comparison of local wall-thickness estimators.

## 3.1. Overview

The input to the method is what a clinical short-axis (SAX) cardiac MRI examination provides once the left ventricle (LV) has been outlined, or segmented, on each image. In a SAX acquisition the heart is imaged in a stack of parallel planes cut perpendicular to the long axis of the ventricle. On each of these roughly ten planes the segmentation yields one closed ring of points on the endocardium, the inner boundary of the blood-filled cavity, and one on the epicardium, the outer boundary of the muscular wall, which is called the myocardium. Each stack is labelled as acquired at end-diastole (ED), when the ventricle is fullest, or at end-systole (ES), when it is most contracted. Panel (b) of Figure 3.1 shows one segmented SAX image, and panel (d) the stacks of rings obtained from all the planes of one ACDC case at ED and ES. The planes are several millimetres apart, so the wall is observed only where a plane cuts it and has to be inferred everywhere else. The required outputs are the two closed surfaces that pass through these rings and fill the gaps between them (panel (e)), a thickness value at every point of the reconstructed wall (panel (f)), and a summary of those values in the 17-segment layout of the American Heart Association (AHA), the standard division of the ventricle used in clinical reporting (panel (g)).

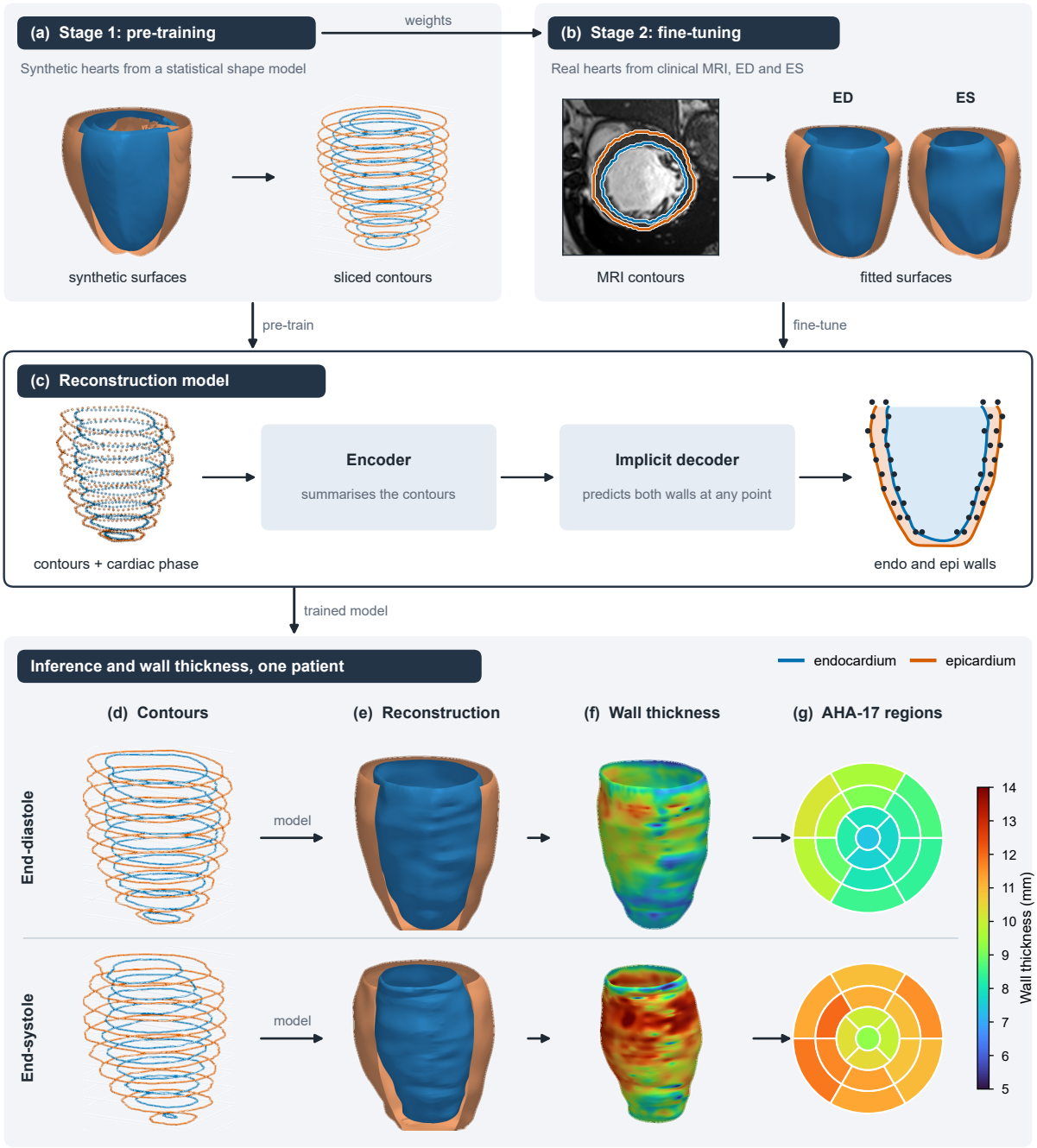



**Figure 3.1.** Overview of the methodology. (a) Stage 1 pre-trains the model on synthetic ED surfaces sampled from the statistical shape model and on the SAX contours sliced from them. (b) Stage 2 fine-tunes the same model, starting from the Stage 1 weights, on clinical ED and ES cases, whose contours come from expert SAX labels and whose target surfaces are smooth fits to those contours. (c) The model reads the contour points together with the cardiac phase and returns the endocardial and epicardial surfaces; the long-axis section shows the reconstructed walls of one ED case, with dots where the observed rings cross the section plane. (d)–(g) Inference on one ACDC case at ED (top) and ES (bottom): observed SAX contours, reconstructed surfaces with the epicardium partly cut away, local wall thickness measured with the Laplace field method, and its mean over each AHA-17 segment, with the anterior wall at the top and the septum on the left. Both phases share one colour scale, and surfaces are smoothed for display only. The case in (b)–(g) illustrates the pipeline; cohort results are reported in Chapter 4.

Three obstacles shape the methodology. The first is that no public dataset pairs such contour stacks with independently acquired three-dimensional LV surfaces. The examples the model needs to learn

from do not exist and had to be constructed. Three-dimensional LV surfaces were therefore built from two sources. The first is a statistical shape model (SSM), a compact description of how ventricle shape varies across a population. Sampling it gives complete surfaces, which are used to pre-train the model, that is, to teach it the general shape of a ventricle before it sees any patient (panel (a)). The second source is the clinical contours of two public datasets, ACDC and M&Ms-2, through which smooth surfaces are fitted. These give patient-specific ED and ES targets for fine-tuning, the second training stage, in which the model adapts to real patients (panel (b)).

The second obstacle is that the two surfaces are not independent: the epicardium must lie outside the endocardium everywhere, including between the slices where nothing is observed. The model therefore represents the endocardium by a signed-distance field (SDF), a function that gives at every point in space the distance to the surface, negative inside the cavity and positive outside. The epicardial field is obtained by subtracting a strictly positive offset from it, so the two surfaces cannot cross. The model also receives the cardiac phase as an input, so a single model can represent ED and ES.

The third obstacle is that wall thickness has no unique definition on a curved wall: the value depends on how a point on the inner surface is paired with a point on the outer surface. Instead of fixing one definition in advance, four established estimators are run on the same reconstructed surfaces, the most consistent one is selected, and only then are regional AHA-17 values computed.

Most of the building blocks are established methods, and the chapter names their origin where each appears. The model has two parts (panel (c)): an encoder, which reads the contour points and condenses them into a short description, and a decoder, which turns that description into the two fields. The decoder is of the DeepSDF type [13]: given the description of a shape, it returns the signed distance at any point, so the surface can be read out at any resolution. The encoder is a PointNet [34]: it applies the same small network to every point and keeps, for each learned feature, the strongest response, so the result does not depend on how many points there are or in what order they arrive. Because such a summary forgets where each point was, a second encoder keeps a coarse three-dimensional grid of local features, as in Convolutional Occupancy Networks [38]. Two standard ingredients complete the model. Fourier features [16] present the coordinates of a point as sine and cosine waves, which makes fine detail easier to learn, and the regularisation terms of Gropp et al. [15] make the predicted field behave like a true distance. Outside the model, the clinical targets are built with the radial basis function (RBF) surface fitting of Carr et al. [24], a classical way to pass a smooth surface through scattered points; the population shape model is the UK Digital Heart Project atlas of Bai et al. [4]; and the thickness estimators and the AHA-17 layout come from the literature reviewed in Section 2.3.3.

What is new is how these parts are put together to close the gap identified in Section 2.3.4. Every contour point carries its tissue and its cardiac phase, so one model serves ED and ES. Endocardial and epicardial points are summarised separately before they are merged. The epicardial field is tied to the endocardial field by a positive offset, which guarantees a positive wall and is trained directly against the measured wall distance. Training moves from synthetic SSM geometry to clinical ED and ES targets through one shared sample format (cache). And the evaluation is designed for a setting in which no

three-dimensional ground truth exists. Table 3.3 in Section 3.3 summarises this division for the model itself.

The chapter follows the pipeline of Figure 3.1. Section 3.2 constructs the two supervision streams and the shared sample format. Section 3.3 describes the model, and Section 3.4 how it is trained, how surfaces are extracted from it, and which alternative reconstructions it is compared against. Section 3.5 defines the thickness estimators, the three-dimensional thickness maps, and the AHA-17 aggregation. Chapter 4 then applies this protocol: it measures how closely each reconstruction follows the observed rings and how far it agrees with an RBF fit and an SSM fit of the same rings, how the reconstructions degrade as slices are removed, how the four thickness estimators agree with one another, and how regional thickness differs between healthy hearts and hearts with hypertrophic cardiomyopathy, a disease in which the wall thickens abnormally, and between ED and ES. These analyses answer, in turn, the three research questions of Section 1.3.

## 3.2. Data Preparation

The model is trained to map contours to surfaces, so every training case must contain both: a sparse contour stack as input and a pair of closed three-dimensional surfaces as target. Public cardiac MRI datasets provide only the first half. ACDC and M&Ms-2 supply images and slice-wise segmentation masks, label images in which every pixel is marked as cavity, myocardium, or background, but not the closed ("watertight") endocardial and epicardial surfaces that a signed-distance decoder needs as supervision. The training data were therefore constructed from two sources that fill this gap in complementary ways. The UK Digital Heart Project LV SSM[1] supplies complete surfaces over a controlled range of healthy shapes, from which 1300 synthetic ED cases were generated for pre-training. The clinical segmentations supply patient-specific anatomy in both cardiac phases, and were converted into closed surfaces by fitting smooth implicit functions to their contours, for fine-tuning. The two sources are described in turn, followed by the shared format, augmentation, and patient-level partitions through which they enter training.

### 3.2.1. Synthetic ED geometry from the statistical shape model

A statistical shape model describes a population of shapes by a mean shape and a small number of principal modes of variation, the main independent ways in which the shapes of the population differ from that mean [10]. Sampling it yields complete, closed surfaces whose variation is known and controlled, which makes it a convenient source of supervision for pre-training. What it cannot provide matters just as much: the sampled shapes carry no scanner-dependent acquisition effects, no variation in how a clinician would have drawn the contours, and, for the model used here, no systolic geometry. The synthetic stream is therefore used to teach the model what an LV looks like in general, before the clinical stream exposes it to the conditions of the evaluation data.

[1] https://github.com/UK-Digital-Heart-Project/Statistical-Shape-Model

The shape model is the UK Digital Heart Project LV SSM of Bai et al. [4], built from more than 1,000 high-resolution cardiac MR images of healthy subjects. It provides a mean endocardial surface of 22 043 vertices (the corner points of the triangles that make up the mesh) together with 100 principal modes of variation, stored as a $66\,129 \times 100$ matrix of mode vectors, and a corresponding vector of 100 eigenvalues $\lambda_i$, each measuring how much of the population's variation its mode explains. A new endocardial mesh is generated by

$$X(b) = \bar{X} + \Phi b, \tag{3.1}$$

where $\bar{X}$ is the mean shape flattened to a vector, $\Phi$ is the matrix of mode vectors, and $b = [b_1, \ldots, b_{100}]^\top$ is the sampled coefficient vector. The standard deviation $\sigma_i = \sqrt{\lambda_i}$ sets the scale of variation along mode $i$. The first modes capture the largest sources of variation in the population: mode 1 mainly controls overall LV size, mode 2 the elongation along the long axis, and higher modes increasingly local deformations (Figure 3.2).

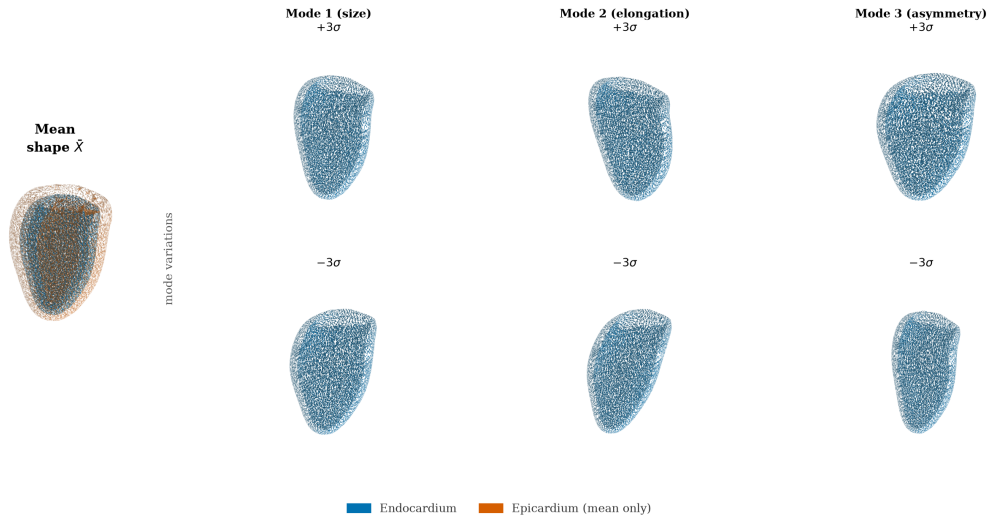



**Figure 3.2.** UK Digital Heart Project LV SSM mean shape and first three PCA modes at $\pm 3\sigma$.

Of the 100 modes, the first 32 capture 99% of the shape variance and define the sampling space. Each candidate coefficient vector $b$ was drawn by sampling each active component as $b_i \sim \mathcal{N}(0, \sigma_i^2)$ and was accepted only if it satisfied

$$\left|\frac{b_i}{\sigma_i}\right| \le 3, \qquad \sum_i \left(\frac{b_i}{\sigma_i}\right)^2 \le \chi^2_{0.99,d}, \tag{3.2}$$

where $d$ is the number of active modes and $\chi^2_{0.99,d}$ is the 99th percentile of the chi-squared distribution with $d$ degrees of freedom. The first condition keeps every individual mode within three standard deviations of the mean. The second bounds the overall deviation: the sum measures how far the sample lies from the mean shape once every mode is weighed by its own spread (the Mahalanobis distance), and the threshold admits only combinations that would occur in 99% of the population. Each accepted vector is substituted into Equation (3.1) to obtain new vertex positions, which are connected into triangles exactly as in the mean mesh; the result is a closed and consistently oriented surface by construction. Each surface is then checked against four simple geometric descriptors, namely its enclosed volume,

its surface area, a sphericity index (how close the shape is to a sphere), and the direction of its surface normals (the directions perpendicular to the surface) at the apex and base, and it is discarded if any value falls outside bounds calibrated on the population; exact volume and area computations on every fortieth sample confirmed that the fast descriptors were reliable. Of 2048 candidates, 1300 (63%) were accepted.

The SSM provides only the endocardial surface. An epicardial surface was synthesised by moving each endocardial vertex $p_i$ outward along its normal $n_i$ by a distance $d_i$,

$$q_i = p_i + d_i n_i, \tag{3.3}$$

with $d_i$ set to approximately 10 mm near the base and 5 mm near the apex, plus Gaussian noise with a standard deviation of 0.5 mm so that the wall is not perfectly uniform. This synthetic epicardium exists only to give the decoder a consistent outer surface during pre-training, so that the model learns the general relation between an inner and an outer LV surface before it sees clinical data. It does not reproduce clinical wall thickness and is never used as a reference for any thickness result.

Finally, each surface pair was converted into the form the model will see. Slicing both surfaces at 10 evenly spaced levels along the long axis produces sparse SAX contour rings that mimic a clinical acquisition. For each case, 2048 query points, locations in space at which the decoder will be asked for a value during training, were sampled near the surfaces and in the surrounding volume, and the signed distance to each surface, or an inside/outside label, was recorded at every one of them; these points provide the dense volumetric supervision for the decoder. Figure 3.3 summarises the whole procedure, from the accepted region of coefficient space to the contour rings and query points of one case. The synthetic stream fixes the general shape of the problem; the clinical stream, described next, supplies the anatomy on which the model is evaluated.

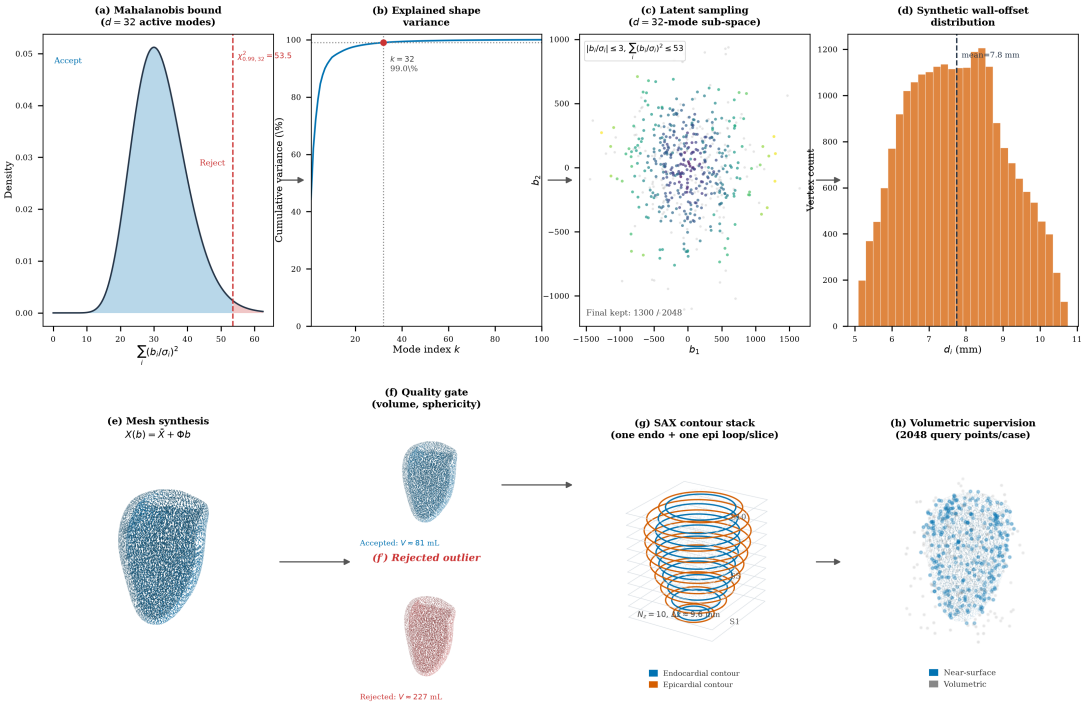



**Figure 3.3.** Synthetic ED dataset generation, from SSM sampling to geometric supervision. Top row, statistical constraints: (a) distribution of the overall-deviation (Mahalanobis) statistic of Equation (3.2) with the $\chi^2_{0.99,32}$ acceptance threshold; (b) cumulative shape variance, with 32 active modes reaching 99%; (c) accepted coefficient samples in the first two modes; (d) distribution of the synthetic epicardial offset of Equation (3.3). Bottom row, geometric construction: (e) mesh synthesis via Equation (3.1); (f) an accepted shape next to a rejected outlier; (g) the 10 SAX contour levels; (h) the 2048 near-surface and volumetric query points used for decoder supervision.

### 3.2.2. Clinical ED and ES surfaces from SAX labels

The clinical data come from two public collections, the Automated Cardiac Diagnosis Challenge (ACDC) [7] and the second Multi-Centre, Multi-Vendor and Multi-Disease Cardiac Segmentation challenge (M&Ms-2) [19], which together provide 460 patients (100 from ACDC and 360 from M&Ms-2) with outlines of the cavity, the myocardium, and the right ventricle drawn by experts at ED and ES. Both are available for research at no cost after registration and acceptance of a data-use agreement, and the shape model of the previous section is likewise freely available. These data contribute what the synthetic stream lacks: scanner and site variability, the way clinicians actually delineate the wall, patient-specific and in part pathological anatomy, and the contracted geometry of end-systole.

Segmentation masks are not yet surfaces. Figure 3.4 shows representative ACDC slices with the cavity, myocardium, and right-ventricular labels, and the ED labels of one patient stacked at their true through-plane positions. The labels are defined one slice at a time, and the slices are often several millimetres apart, so stacking them gives a set of sparse rings with nothing in between. Turning these rings into closed surfaces is a reconstruction problem in itself, and the method chosen for it determines what the model is trained to reproduce.

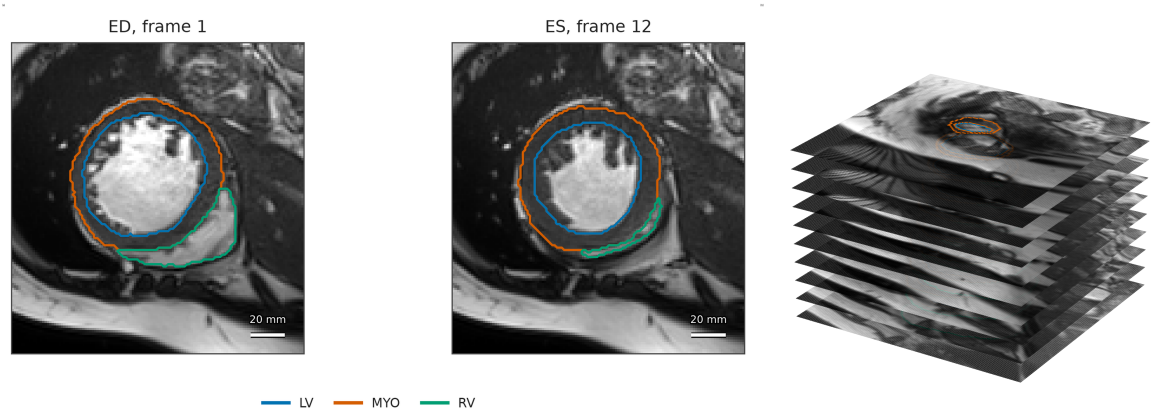



**Figure 3.4.** Clinical-input view used in the real-data pipeline: (a) representative ACDC ED/ES SAX slices with labels and (b) the corresponding sparse ED SAX stack in physical three-dimensional spacing.

Three options were available. Connecting corresponding points on consecutive rings (lofting) requires a one-to-one pairing of vertices between rings that the data do not provide, and gives a surface that is only piecewise linear between slices. Fitting the SSM would impose a population prior on every target and, because the available model has ED modes only, would either force systolic hearts into a diastolic shape space or treat the two phases differently. Fitting a smooth implicit function to the rings has neither drawback: it needs no vertex pairing and no atlas, and the same procedure applies to ED and ES alike. The clinical targets were therefore built by Radial Basis Function (RBF) fitting, which leaves the SSM in its role as the source of synthetic pre-training geometry.

Each segmentation is distributed as a NIfTI file, the standard container for medical image volumes. The file stores the label volume together with the physical size and position of its voxels, the three-dimensional pixels of the volume. The volume is first reoriented to a common anatomical convention, called RAS+, in which the three axes point towards the patient's right, anterior, and superior sides; the physical voxel spacing is preserved, so that every patient is expressed in the same frame and in millimetres. Each voxel of the label volume carries an integer that names the structure it belongs to, such as cavity, myocardium, or right ventricle; because the two datasets number these structures differently, the labels are mapped to a common definition of cavity and myocardium. On every acquired SAX slice, the boundary of the cavity gives the endocardial ring and the outer boundary of cavity plus myocardium gives the epicardial ring; when a slice contains more than one closed boundary, for instance because of a small detached fragment of the label, only the largest ring is kept. Each ring is resampled to 60 evenly spaced points, so that every slice contributes the same number of points regardless of image resolution, and the points are stored at the measured position of the slice along the long axis. These points serve two purposes: they are the sparse observation the encoder receives, and they are the constraints from which the target surfaces are built.

The endocardial and epicardial surfaces are fitted separately. For either surface $s$, let $p_i$ be a contour point and $n_i$ its outward in-plane normal, estimated from the local contour tangent and pointing away from the ring centre. Following Carr et al. [24], a smooth function $f_s$ is defined over the whole volume as a sum of radial basis functions (RBFs), smooth bumps whose value depends only on the distance to a

centre $c_j$, plus a linear term, and the surface is taken as its zero level set, the set of points where the function is zero. With the thin-plate-spline kernel $\phi$ used here, the function is

$$f_s(x) = \sum_{j=1}^{K} w_j \phi(\|x - c_j\|_2) + a_0 + a^\top x, \qquad \phi(r) = r^2 \log r, \tag{3.4}$$

where the centres $c_j$ are the constraint points and the weights $w_j$, $a_0$, and $a$ are chosen so that the function satisfies the constraints. The contour points fix where the function is zero, and paired points placed a distance $d = 2.5$ mm inside and outside each contour point fix its sign and local scale:

$$f_s(p_i) = 0, \qquad f_s(p_i + dn_i) = d, \qquad f_s(p_i - dn_i) = -d. \tag{3.5}$$

A regularisation parameter of 0.5 lets the function pass close to, rather than exactly through, the constraints, which reduces its sensitivity to small irregularities in the drawn contours.

Each fitted function is evaluated on a 1.0 mm grid that extends 8.0 mm beyond the rings in-plane and 1.5 mm beyond the first and last ring along the long axis, so that the surface can close just beyond the observed extent. The region where $f_s \leq 0$, the inside of the surface, is reduced to its valid connected part (small disconnected pockets are discarded) and converted to a signed-distance volume, and the volume is smoothed with a Gaussian filter of width 1.2 mm, which replaces each value by a weighted average of its neighbours, before marching cubes, a standard algorithm that extracts a triangle mesh from the zero level set on a grid, produces the surface [14]. Smoothing the field at this stage removes the small step-like ripples that the voxel grid would otherwise imprint on the surface, without moving mesh vertices afterwards. The extracted mesh is accepted unchanged if it is a single closed shell with consistently oriented faces and no handles or tunnels; otherwise only the failed property is repaired. Cases with fewer than two valid rings per surface, a failed fit, a surface that could not be repaired into a single closed shell, implausible dimensions, or an endocardium not fully enclosed by its epicardium are excluded. The full sequence of operations and their parameters is listed in Table A.2 of Chapter A, and Figure 3.5 illustrates it on one case: the endocardial and epicardial constraints are fitted separately, and their surfaces then pass through the same checks.

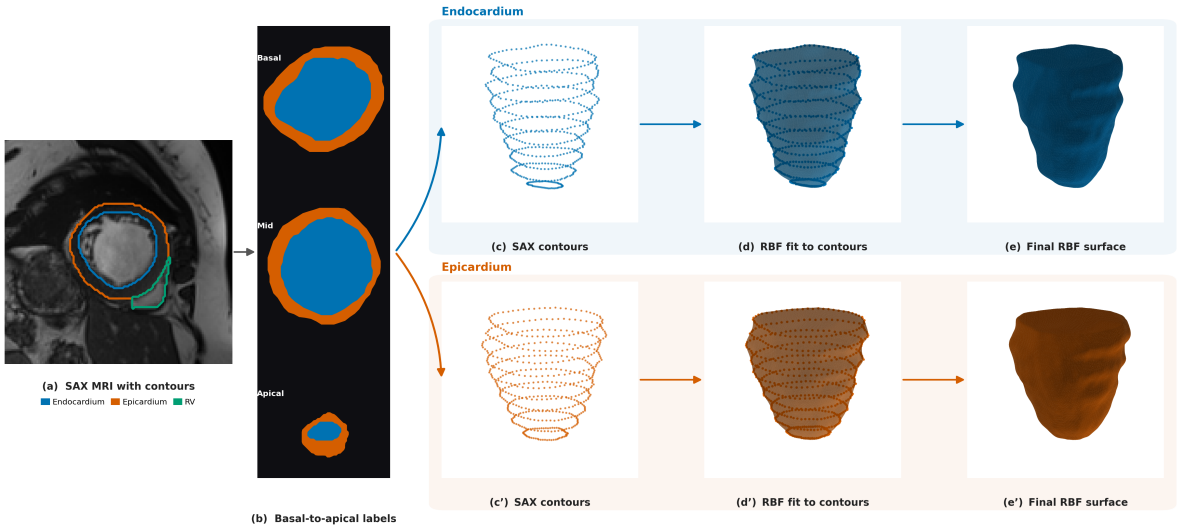



**Figure 3.5.** Clinical RBF contour-to-surface pipeline. Panel (a) shows one acquired SAX slice with the expert endocardial, epicardial, and right-ventricular contours. Panel (b) shows the corresponding basal, mid-cavity, and apical labels, while panels (c) and (c′) stack the extracted contour rings. Panels (d) and (d′) overlay those constraints on the zero-level surfaces extracted from the smoothed RBF fields, and panels (e) and (e′) show the final topology-checked surfaces used to construct the clinical supervision.

The two meshes of each case are centred and scaled with the same translation and scale factor, and 2048 query points are drawn near the surfaces and throughout a box that encloses them with a margin, with their signed distances recorded, exactly as for the synthetic stream.

One property of these targets governs the interpretation of every later result. They are *surrogate targets*: RBF-derived surfaces that stand in for the three-dimensional reference geometry that does not exist, and whose shape between the slices depends on the kernel, the off-surface constraints, the regularisation, and the extraction resolution just described. Agreement between the model and these targets shows that the model has learned to reproduce a particular, reasonable reconstruction of the rings, not that it has recovered the true anatomy between them. Chapter 4 keeps this distinction explicit by also measuring every reconstruction against the observed rings themselves.

### 3.2.3. Shared sample format, augmentation, and splits

Both streams enter training through the same sample format, the same augmentation rule, and partitions defined at the level of patients.

Every case, synthetic or clinical, is stored as one item with the fields listed in Table 3.1: the contour points with their tissue and phase labels, surface samples and query points derived from the target meshes, and the coordinate information that keeps observation and target in the same frame. The ED and ES items contain 10 SAX levels, 60 points per available ring, and 2048 query points per case. The format deliberately separates what the model observes from what it learns from. The encoder receives only the first three fields; the decoder is supervised by the remaining ones, which are never shown to the encoder. At inference time the model therefore needs nothing but the contours, and the architecture and loss described in the next two sections do not change when training moves from the synthetic to the clinical stream.

**Table 3.1.** Fields of one stored sample, shared by the synthetic and clinical streams, and their role in training.

| Field | Origin | Role |
|---|---|---|
| Contour points | Synthetic mesh slicing or clinical slice labels | Sparse SAX observation for the encoder |
| Tissue label | Contour extraction | Distinguishes endocardial from epicardial points during tissue-specific pooling |
| Phase label | Dataset metadata | Conditions the encoder on ED or ES geometry |
| Surface samples | Prepared endocardial and epicardial meshes | Surface-fitting and wall-offset supervision |
| Query points and targets | Near-surface and volumetric mesh sampling | Dense signed-distance and off-surface supervision |
| Mesh metadata | Preprocessing pipeline | Coordinates, scale, faces, and orientation for evaluation and visualisation |

The clinical stream is small, and its contours vary in how they were drawn and acquired. To make the encoder robust to such variation, the observed contours are perturbed during training while the targets are left untouched. For each training sample, the data loader applies small independent in-plane shifts to each slice, moving the endocardial and epicardial rings of a slice together so that their relation is preserved, as well as small random displacements of individual points, a global change of scale, and random removal of contour points; with lower probability it also drops whole slices, always keeping at least three, or rotates the stack about the long axis. The target mesh, surface samples, query points, and tissue and phase labels stay fixed, so the encoder must recover one fixed target from many plausible versions of its sparse observation (Figure 3.6). Such label-preserving perturbations are a standard way to improve robustness when labelled data are limited [39, 40]. Augmentation is applied to both streams and only after the data have been split, so that no augmented view of a test patient is ever seen in training.

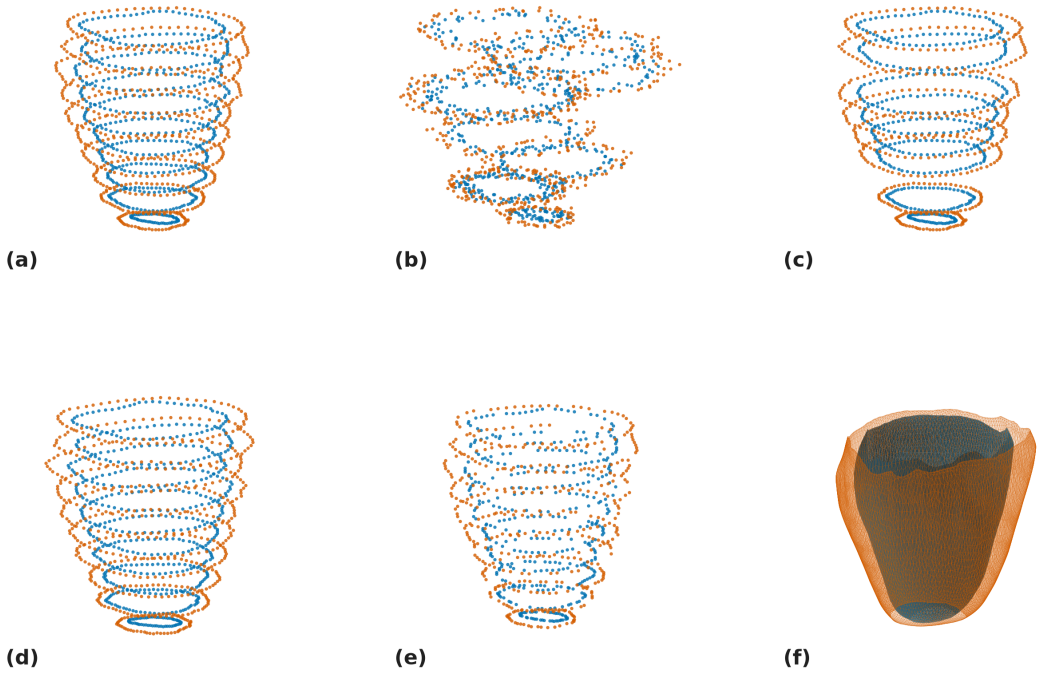



**Figure 3.6.** Examples of contour-input augmentation: the unaltered contour stack, slice translation, slice rotation, and contour-point dropout. Endocardial and epicardial points are shown in blue and orange, respectively; the target mesh is the same in every case.

Not every clinical patient yields a usable target. A study may contain only long-axis views or too few slices, or its segmentation may fail the plausibility checks of Section 3.2.2. After these checks, 444 of the 460 patients provide a valid ED mesh and 423 a valid ES mesh; together with the 1300 accepted synthetic meshes they form the three streams of Table 3.2. The clinical patients are divided into training, validation, and test sets in a 70 %/15 %/15 % proportion, rounded to whole patients and applied to patients rather than to meshes, so that the ED and ES meshes of one patient always fall in the same set and no information passes from the test patients into training. The synthetic meshes are used for pre-training only and never appear in the validation or test sets, so every number reported in Chapter 4 is computed on clinical cases. The role of each stream in the two training stages is summarised in Chapter A. With every case in one format, the model can now be described without reference to the source of a sample.

**Table 3.2.** Final sample counts and partitioning of the three data streams. Real streams report patients that pass the quality checks; the synthetic stream reports meshes accepted out of the 2048 sampled shape vectors. The reported clinical evaluation contains no synthetic cases.

| Stream | Source | Phase | Count |
|---|---|---|---|
| Real data | ACDC and M&Ms-2 | ED | 444 patients |
| Real data | ACDC and M&Ms-2 | ES | 423 patients |
| Synthetic data | UK Digital Heart SSM | ED | 1300 meshes |

## 3.3. Model Architecture

The data impose three requirements on the model. Its input is a set of a few hundred points whose number varies from case to case and whose order carries no meaning, so the encoder must be indifferent

to point count and order. Its output is a pair of surfaces that must be continuous between the slices and correctly nested everywhere, so the decoder must produce two related fields rather than one. And the same model must represent diastolic and systolic hearts, so the phase must reach every stage that interprets the contours. Figure 3.7 shows the resulting design. A global encoder summarises the whole contour set in one vector, a local encoder records where in space the contours lie, and an implicit decoder, evaluated at any point, combines both with the coordinates of that point to return the endocardial signed distance and the positive offset that defines the epicardial field.

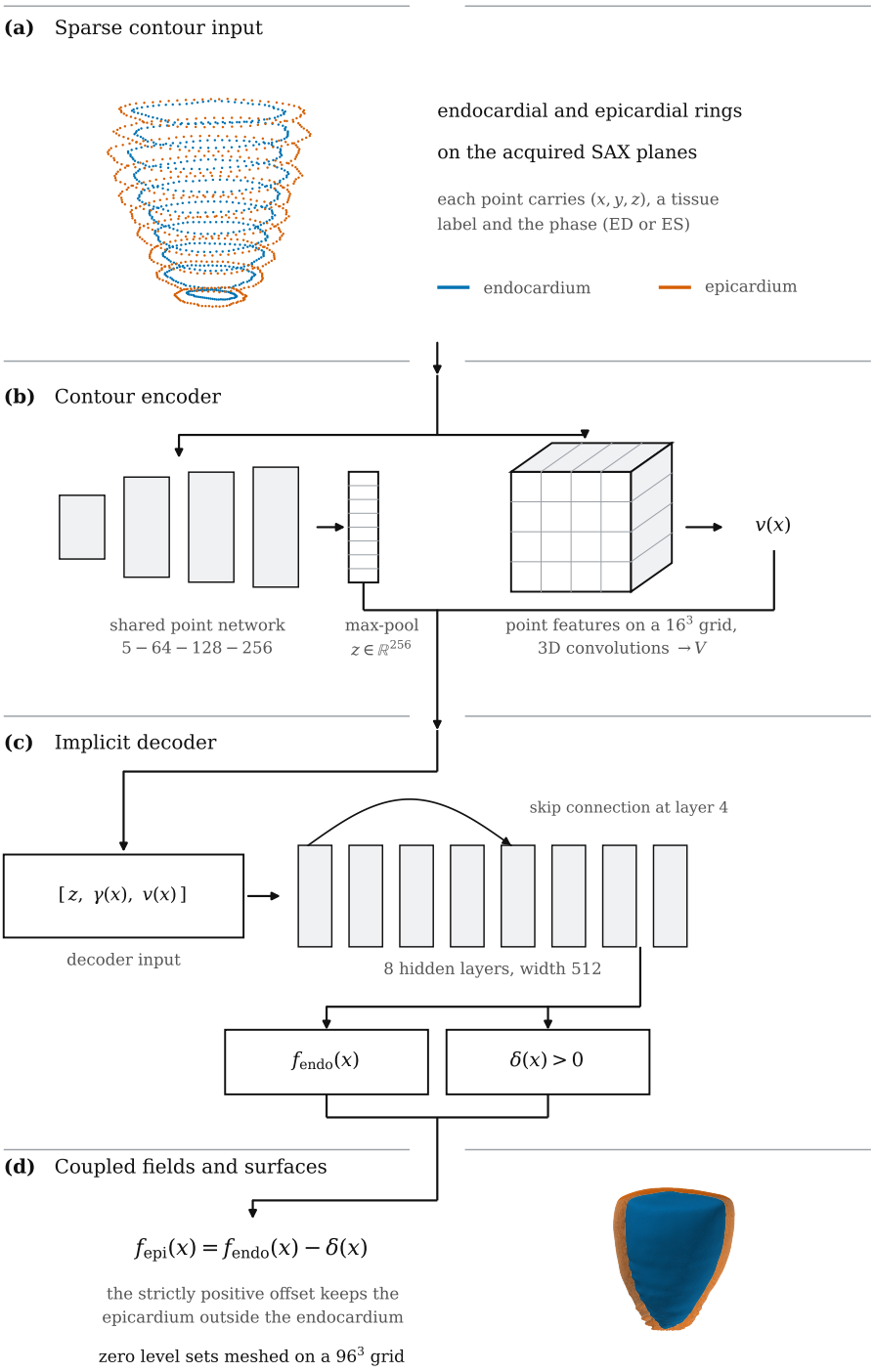



**Figure 3.7.** Phase-conditioned model architecture, from the sparse contour observation (a) through the global and local encoder paths (b) and the implicit decoder (c) to the coupled fields and extracted surfaces (d). The point cloud in (a) and the cut surfaces in (d) are the observed contours and the corresponding reconstruction of one ACDC case, with the epicardium opened to expose the endocardial cavity. The offset $\delta$ shown in the figure is a separation between field values, not a measured thickness.

Each of these parts has a precedent, and Table 3.3 separates what is borrowed from what is specific to this work. The decoder is a signed-distance network conditioned on a latent code, a compact vector of numbers that summarises one shape, as introduced by DeepSDF [13]; here, however, the code is produced by an encoder from the observed contours rather than fitted separately to every shape by optimisation, so that a new case is reconstructed in a single evaluation of the network. The global

encoder is a PointNet [34], in which the same small network is applied to every point and the results are combined by max-pooling, keeping for each feature the largest value over all points; this is the standard way to encode a set of points whose order carries no meaning. The local feature volume follows Convolutional Occupancy Networks [38], which showed that a single summary vector loses the information of where each piece of evidence was, and that spreading the point features over a coarse three-dimensional grid and reading that grid near each query point restores it. The positional encoding is the Fourier-feature mapping of Tancik et al. [16]. Compared with the recent cardiac implicit models reviewed in Section 2.3.2, which reconstruct from image data [9] or from sparse segmentations of both ventricles with long-axis views and a learned biventricular prior [29], the proposed model takes only LV contour points with a tissue and a phase label, and it makes the relation between the two surfaces part of the architecture: the epicardial field is defined from the endocardial field through a positive offset that is supervised directly.

**Table 3.3.** Model components, the prior work each one builds on, and what is specific to this work.

| Component | Builds on | Specific to this work |
|---|---|---|
| Input representation | Unordered point-set input [34] | Every point carries a tissue label and the cardiac phase, so one model serves ED and ES |
| Global encoder | Shared per-point network with max-pooling [34] | Endocardial and epicardial points are pooled separately before being merged into the latent code |
| Local feature volume | Convolutional feature grid read at the query point [38] | Built from a few hundred contour points averaged into a coarse $16^3$ grid; fed to the decoder trunk and to the offset head |
| Implicit decoder | Latent-conditioned signed-distance network [13] with Fourier features [16] | Latent code supplied by the contour encoder instead of per-shape optimisation |
| Output coupling | Surfaces as zero level sets of signed-distance fields [13] | Epicardial field defined as the endocardial field minus a strictly positive offset $\delta$, which a wall-distance loss supervises directly |
| Training | Surface and distance fitting [13]; Eikonal regularisation [15] | Two stages, synthetic SSM pre-training then clinical ED/ES fine-tuning, through one shared sample format |

### 3.3.1. What the model observes

For a batch of $B$ cases processed together, the encoder receives the contour points as one padded array, or tensor,

$$P \in \mathbb{R}^{B \times N \times 5} \tag{3.6}$$

and a validity mask $M \in \{0,1\}^{B\times N}$ that marks which of the $N$ rows are real points; padding only allows cases with different numbers of points to share a batch. Each valid row is $p_i = (x_i, y_i, z_i, t_i, q_i)$: normalised spatial coordinates, a tissue label $t_i$ that distinguishes endocardial from epicardial points, and a phase label $q_i$ that identifies ED or ES. The network receives neither image intensities nor a template mesh; everything it knows about a case comes from these five values per point. Attaching the phase to every point, rather than introducing it only at the decoder, lets both encoders interpret the same arrangement of contours differently for a diastolic and a systolic heart. The tissue label serves a different purpose: it allows evidence about the cavity boundary and evidence about the outer boundary to be summarised separately before they are combined.

### 3.3.2. Contour encoder

The global pathway is a PointNet [34]: a shared network with layer widths $5 \to 64 \to 128 \to 256 \to 256$, with ReLU activations (simple non-linear functions) between the layers, processes every point independently, so that no ordering is imposed within a ring or between slices, and the per-point features are then pooled. Writing $P = \{p_i\}_{i=1}^{N}$ for the points of one case, each point receives the 256-dimensional feature

$$h_i = \phi_{\text{enc}}(p_i), \tag{3.7}$$

where $\phi_{\text{enc}}$ is the shared network. The departure from the standard PointNet is in the pooling. Instead of one maximum over all points, the maximum is taken separately over the endocardial set $\mathcal{E}$ and the epicardial set $\mathcal{P}$,

$$z_{\text{endo}} = \max_{i\in\mathcal{E}} h_i, \qquad z_{\text{epi}} = \max_{i\in\mathcal{P}} h_i, \tag{3.8}$$

independently in every feature channel, so that the strongest response to each learned geometric pattern is kept for each boundary before the two are merged. The two descriptors are concatenated into a 512-dimensional vector and projected to the global latent code

$$z = W_z[z_{\text{endo}}, z_{\text{epi}}] + b_z, \tag{3.9}$$

with $z \in \mathbb{R}^{256}$, which gives the decoder a compact description of the whole case.

Max-pooling has a cost: compressing all points into one vector discards where each piece of evidence was located. The local pathway restores this information, following Convolutional Occupancy Networks [38]. A second shared network with widths $5 \to 64 \to 64 \to 32$ assigns a 32-dimensional feature to every valid point, and the features are scattered into a regular $16^3$ grid covering the normalised extent $[-1.8, 1.8]^3$; the features of points that fall in the same cell are averaged, so the content of a cell does not depend on how many contour points it happens to contain. Three $3\times3\times3$ convolutions, small filters that combine each cell with its immediate neighbours, with Group Normalisation (a rescaling of the features that stabilises training) and ReLU after the first two, spread information into neighbouring cells and produce

$$V \in \mathbb{R}^{32\times16\times16\times16}. \tag{3.10}$$

For a query point $x$, the local feature is read from this volume by trilinear interpolation, a weighted average of the eight surrounding cells,

$$v(x) = \mathsf{Interp}(V, x). \tag{3.11}$$

Unlike $z$, which is the same for every query of a case, $v(x) \in \mathbb{R}^{32}$ changes with the query location and describes the contour evidence near it. The grid is deliberately coarse: a few hundred contour points cannot support a fine volume, and its role is to give context to the decoder, not to serve as the reconstruction grid.

### 3.3.3. Implicit decoder with positive-offset coupling

The decoder represents each surface as the zero level set of a function of continuous coordinates, in the manner of DeepSDF [13]. A network fed with plain coordinates tends towards overly smooth fields, so each query point is first expanded with Fourier features [16], which add sine and cosine components at several spatial frequencies. For $x \in \mathbb{R}^3$, the encoding is

$$\gamma(x) = [x, \mathsf{sin}(2^0\pi x), \mathsf{cos}(2^0\pi x), \ldots, \mathsf{sin}(2^{L-1}\pi x), \mathsf{cos}(2^{L-1}\pi x)], \tag{3.12}$$

with $L = 3$, giving $3 + 6L = 21$ components. The encoded coordinate is concatenated with the global code to form the decoder input

$$h_{\mathsf{in}}(x) = [z, \gamma(x)] \in \mathbb{R}^{277}. \tag{3.13}$$

An input layer maps this vector to width 512, and the main body of the decoder, its trunk, applies eight fully connected layers of the same width with Softplus activations, smooth non-linear functions that keep the field smooth and differentiable. At the fourth hidden layer, the 277-dimensional input is appended to the current state and projected back to width 512; this skip connection lets the deeper layers see the case code and the coordinate directly rather than through the whole trunk. The local feature $v(x)$ enters the decoder at the input layer, at this skip connection, and in the offset head described next, so that the prediction at a point depends on the nearby contours as well as on the global shape.

The output stage is where the model departs from an ordinary implicit decoder. Predicting the endocardium and the epicardium as two unrelated fields would let them cross or touch wherever the data are sparse, producing a wall of zero or negative thickness between slices. The final hidden representation therefore feeds two coupled output layers, or heads: one predicts the endocardial signed-distance value $f_{\mathsf{endo}}(x)$, the other a strictly positive offset $\delta(x)$, and the epicardial field is defined as

$$f_{\mathsf{epi}}(x) = f_{\mathsf{endo}}(x) - \delta(x), \qquad \delta(x) > 0. \tag{3.14}$$

Positivity is enforced in the implementation by a sigmoid-based mapping, which squashes any real number into a bounded positive range, with additional headroom for large offsets. Because the epicardial field is always lower than the endocardial field by $\delta$, its zero level set always lies outside the endocardial one, which rules out wall collapse by construction. Two qualifications apply. The constraint does not by itself guarantee an anatomically valid surface or a valid mesh once the field has been sampled on a grid, which is why the extracted meshes are checked and repaired in Section 3.4.2. And $\delta$ is a difference between two field values at one point, not the distance between the two extracted surfaces, so it is not

a measurement of wall thickness; thickness is measured on the extracted surfaces with the methods of Section 3.5. How the two fields are turned into meshes is described in Section 3.4.2; the complete configuration of the model is listed in Table A.4 of Chapter A.

## 3.4. Training and Inference

Training proceeds in two stages, pre-training on the synthetic SSM cases and fine-tuning on the clinical ED and ES cases, with the same architecture and the same loss throughout. This section describes the two stages and the loss, then how the trained fields are turned into meshes and an offset map, and finally the two alternative reconstructions against which the model is compared.

### 3.4.1. Two-stage training and loss

In the first stage the model is trained only on the 1300 synthetic ED cases, which give it a broad prior for LV shape under controlled conditions and tolerate stronger augmentation of the sparse observations. In the second stage it continues from the pre-trained weights on the ACDC and M&Ms-2 cases of both phases, adapting to clinical contour quality and to systolic anatomy. The patient-level split and the augmentation rule of Section 3.2.3 apply unchanged.

One phase-conditioned model is trained rather than one model per phase. The reason follows from the data. Because the SSM provides ED geometry only, a separate ES model could not have received equivalent pre-training, whereas a shared model lets the ES cases benefit from the prior learned on the synthetic ED corpus, with the phase label telling the encoder which configuration it is seeing. The RBF-derived targets are constructed identically in both phases, so no second asymmetry is introduced at fine-tuning. The cost is that phase information is coarse, a binary label rather than a position in the cardiac cycle, and that the prior remains biased towards diastolic shape.

Training adjusts the weights of the network so as to minimise a loss function, a single number that measures how far the current predictions are from what the targets require. Here that requirement has five parts, and each becomes one term of the loss. The predicted surfaces should pass through the sampled points of the target surfaces (surface term). The predicted fields should take the stored signed-distance values at the query points throughout the volume, not only on the surfaces (dense SDF term). Each field should behave like a true distance, changing by one unit per unit travelled, which is expressed by requiring its gradient to have unit length (Eikonal term). The fields should not approach zero anywhere other than at the two surfaces, so values in free space are pushed away from zero (off-surface term). And the offset $\delta$ at the endocardial surface should equal the actual distance to the epicardial target, so that the coupling learns the real wall separation rather than merely keeping the fields ordered (wall term). The first four are standard in implicit surface learning, following DeepSDF [13] and the Eikonal regularisation of Gropp et al. [15]; the fifth is specific to the coupled output.

Formally, let $\mathcal{E}$ and $\mathcal{P}$ denote sampled endocardial and epicardial surface points, $\mathcal{Q}$ the stored query points, $\mathcal{R}$ the union of near-surface and free-space points used for the gradient term, and $\mathcal{F}$ the free-space subset. At each $q \in \mathcal{Q}$, $y_{\text{endo}}(q)$ and $y_{\text{epi}}(q)$ are the stored signed-distance targets, and at each $x \in \mathcal{E}$ the wall target $t(x) = -y_{\text{epi}}(x)$ is the distance from that endocardial point to the epicardial surface.

Writing $\mathbb{E}$ for the mean over the indicated set of points, the terms are

$$\begin{aligned}
\mathcal{L}_{\text{surf}} &= \mathbb{E}_{x\in\mathcal{E}}[|f_{\text{endo}}(x)|] + \mathbb{E}_{x\in\mathcal{P}}[|f_{\text{epi}}(x)|],\\
\mathcal{L}_{\text{SDF}} &= \mathbb{E}_{q\in\mathcal{Q}}[|f_{\text{endo}}(q) - y_{\text{endo}}(q)|] + \mathbb{E}_{q\in\mathcal{Q}}[|f_{\text{epi}}(q) - y_{\text{epi}}(q)|],\\
\mathcal{L}_{\text{eik}} &= \mathbb{E}_{x\in\mathcal{R}}[(\|\nabla_x f_{\text{endo}}(x)\|_2 - 1)^2] + \mathbb{E}_{x\in\mathcal{R}}[(\|\nabla_x f_{\text{epi}}(x)\|_2 - 1)^2],\\
\mathcal{L}_{\text{off}} &= \mathbb{E}_{x\in\mathcal{F}}[e^{-\alpha|f_{\text{endo}}(x)|}] + \mathbb{E}_{x\in\mathcal{F}}[e^{-\alpha|f_{\text{epi}}(x)|}],\\
\mathcal{L}_{\text{WT}} &= \mathbb{E}_{x\in\mathcal{E}}[\ell_{\beta_{\text{WT}}}(\delta(x) - t(x))],
\end{aligned} \tag{3.15}$$

where $\alpha = 5$ and $\ell_\beta$ is the Smooth-L1 penalty, which grows like a squared error for small differences and like an absolute error for large ones, so that a few badly wrong points do not dominate the term:

$$\ell_\beta(a) = \begin{cases} a^2/(2\beta), & |a| < \beta,\\ |a| - \beta/2, & |a| \geq \beta, \end{cases} \qquad \beta_{\text{WT}} = 0.02. \tag{3.16}$$

All distances are expressed in the normalised coordinate system of each case, and the complete objective is

$$\mathcal{L}_{\text{total}} = \lambda_{\text{surf}}\mathcal{L}_{\text{surf}} + \lambda_{\text{SDF}}\mathcal{L}_{\text{SDF}} + \lambda_{\text{eik}}\mathcal{L}_{\text{eik}} + \lambda_{\text{off}}\mathcal{L}_{\text{off}} + \lambda_{\text{WT}}\mathcal{L}_{\text{WT}}. \tag{3.17}$$

where the weights $\lambda$ balance the five terms. The Eikonal term is computed in full precision to keep its gradient calculation stable. Table A.3 in Chapter A summarises the role of each term.

The training run reported here uses the AdamW optimiser, a standard gradient-based method, with a learning rate (step size) of $2 \times 10^{-5}$ and weight decay $5 \times 10^{-4}$; the learning rate follows cosine annealing with warm restarts, decreasing along a cosine curve and being reset at intervals, with an initial restart period of 100 epochs (passes over the training set). Pre-training lasted approximately 200 epochs; fine-tuning then ran for 776 epochs, and the checkpoint at epoch 695 was selected by validation loss. Training uses mixed precision (lower-precision arithmetic where it is safe) and gradient clipping at 1.0 (a cap on the size of each update); each of two GPUs processes a batch of 8 cases and gradients are accumulated over two steps, for an effective batch size of 32. Weights are lightly clipped instead of spectrally normalised, because the decoder is evaluated at many points per step and GPU memory is the limiting resource. Table 3.4 collects these settings.

**Table 3.4.** Training configuration used for the model.

| Setting | Value |
|---|---|
| Training strategy | SSM pre-training, then fine-tuning on real ED/ES |
| Training mode | Combined ED and ES with phase conditioning |
| Augmentation | Contour-input perturbations for synthetic ED and real ED/ES; targets unchanged |
| Optimiser | AdamW |
| Learning rate | $2 \times 10^{-5}$ |
| Scheduler | Cosine annealing with warm restarts ($T_0 = 100$ epochs) |
| Weight decay | $5 \times 10^{-4}$ |
| Training duration | Approximately 200 pre-training epochs, then 776 fine-tuning epochs (best checkpoint: 695) |
| Batch size | 8 per GPU $\times$ 2 GPUs, accumulated for 2 steps (effective batch size 32) |
| Gradient clipping | 1.0 |
| Precision | Automatic mixed precision |
| Hardware | 2 GPUs in parallel |
| Inference grid | $96^3$ |

### 3.4.2. Inference, offset extraction, and comparators

At inference time, the encoder turns the contours into $z$ and $V$ once, and the decoder is evaluated on a regular $96^3$ grid of points. Marching cubes extracts the zero level sets of $f_{\text{endo}}$ and $f_{\text{epi}}$ from this grid as the raw endocardial and epicardial meshes [14]; the mesh is thus a discretisation chosen at inference time, not a property of the architecture, and could be extracted at any resolution. For evaluation, faces of zero area are removed, only the largest connected piece of each mesh is kept, holes are closed, and, where needed, a mesh is rebuilt from a cleaned inside/outside voxel mask. This repair produces usable geometry for surface and thickness measurement, but it also means that the reported metrics describe a repaired mesh rather than the unmodified decoder output.

Because the decoder predicts $\delta$ at every point, a raw offset is also available at every endocardial vertex. In millimetres,

$$o_{\text{raw}}(x) = s\,\delta(x, z, v(x)), \tag{3.18}$$

where $s$ is the scale that maps normalised coordinates back to millimetres. For visualisation only, this offset is calibrated separately for each phase as $o_{\text{cal}} = \alpha\, o_{\text{raw}}$, where $\alpha$ is the ratio of the mean of a reference thickness, computed directly from the segmentation voxels with the distance-transform method of Section 3.5.1, to the mean raw offset. The calibration fixes the mean by construction and is not a test of accuracy, and, for the reason given in Section 3.3.3, the offset itself is not a thickness measurement; every reported thickness value comes from the protocol of Section 3.5.

The model is compared with two reconstructions built from exactly the same contour rings, chosen because they embody two different and well-understood priors. The first is an RBF implicit fit obtained

with the procedure of Section 3.2.2: it interpolates the rings through a smooth field and belongs to the same family as the fine-tuning targets. The second is a fit of the UK Digital Heart Project shape model to the rings. The mean shape is first aligned to the rings by long axis, scale, and rotation about the long axis, where the long axis is the line through the centres of the rings; the fit then alternates three steps, matching each ring point to its nearest model vertex, re-aligning the model to the rings by translation, rotation, and scale, and re-estimating the first $25$ mode coefficients with the coefficients constrained to three standard deviations. As the published model provides ED modes only, this comparator exists at ED alone. Both pass through the same repair and smoothing as the model output.

Since no independent three-dimensional geometry exists, none of the three reconstructions is treated as a reference. They are compared in two ways: by the distance from the observed rings to each surface, which measures how closely each method honours the data it was given, and by pairwise surface and volume agreement, which measures how much the choice of prior changes the result. Neither comparison isolates the contribution of individual model components. The reconstructed meshes now become the input to the measurement stage.

## 3.5. Wall-Thickness Evaluation Protocol

The output of the previous section is a pair of surfaces; the clinical quantity of interest is the distance between them. On a curved, tapering wall that distance is not uniquely defined, because it depends on which point of the epicardium is paired with a given point of the endocardium: the nearest point, the point reached by travelling perpendicular to the surface, or the point reached along a smooth path across the wall. Different pairing rules give different values on the same geometry, and a rule that behaves well on one geometry can be erratic on another. The protocol therefore keeps the two sources of variation apart. Thickness is always measured on the reconstructed endocardial and epicardial meshes; the segmentation volume provides only the contour rings and the orientation of the AHA coordinate system. Four established estimators with different pairing rules are run on the same surfaces, and the one that proves most consistent in Chapter 4 is used for the regional analysis. The protocol compares measurement methods and constructs three-dimensional thickness maps; it does not assess diagnostic performance for infarction, scarring, or hypertrophy.

### 3.5.1. Selected thickness methods

The four methods span the main families identified in Section 2.3.3: distance fields computed on a voxel grid, rays cast from one surface to the other, and methods based on partial differential equations (PDEs), which define a smooth field across the wall and measure along it (Table 3.5). None is assumed in advance to be the best. Each returns a local thickness in millimetres at the endocardial vertices or at points projected onto the endocardium, and the resulting fields are summarised by their mean, median, standard deviation, 5th and 95th percentiles, fraction of valid values, and runtime, and shown as three-dimensional colour maps and as AHA-17 bullseye plots, circular diagrams with one cell per segment.

**Table 3.5.** Wall-thickness methods selected for evaluation on the model-generated LV geometry, with the Euclidean Distance Transform boundary sum kept as the simplest baseline.

| Method | Category | Main measurement principle |
|---|---|---|
| Laplace field | PDE | Transmural field lines from a harmonic potential |
| Yezzi–Prince | PDE | Eulerian transport equations along the transmural field |
| SDF cone rays | Ray / SDF | Median of multiple cone rays around the surface normal |
| EDT boundary sum | Volumetric | Sum of endocardial and epicardial distance transforms |

The Euclidean Distance Transform (EDT) boundary sum is the simplest. A distance transform assigns to every voxel of the myocardium its straight-line distance to the nearest point of a given boundary; the method computes one such transform for each surface and adds them,

$$t^{\mathrm{EDT}}(x) = D_{\mathrm{endo}}(x) + D_{\mathrm{epi}}(x), \tag{3.19}$$

where $D_{\mathrm{endo}}$ and $D_{\mathrm{epi}}$ are the distance transforms of the endocardium and the epicardium. The sum is exact for two parallel flat walls, but it is limited by voxel size and slightly overestimates thickness where the wall is strongly curved, because the two nearest points do not lie on a common path across the wall.

The Laplace field method replaces nearest points with a smooth path across the wall. A scalar field $\psi$ is defined over the myocardium by solving Laplace's equation,

$$\nabla^2 \psi = 0, \qquad \psi|_{\mathrm{endo}} = 0, \qquad \psi|_{\mathrm{epi}} = 1. \tag{3.20}$$

which makes $\psi$ rise as smoothly as possible from 0 on the endocardium to 1 on the epicardium, in the sense that its value at every point is the average of the values around it; a field with this property is called harmonic, and $\psi$ plays the role of a potential whose field lines run across the wall. Thickness is then estimated from the magnitude of the gradient of $\psi$, the rate at which the field changes across the wall, which is small where the wall is thick and large where it is thin. The paths that follow the gradient never cross one another, which is why the method, introduced for the thickness of the brain cortex by Jones et al. [36], is widely used for layered tissue; its weakness is that the numerical estimate can degrade where the wall is thin or the gradient is small. The Yezzi–Prince method [37] removes this weakness by computing, along the same field, the distance travelled from each boundary with a pair of transport equations solved on the voxel grid (an Eulerian formulation, meaning that the quantities are computed at fixed grid points rather than by tracing individual paths), in simplified form

$$\nabla \psi \cdot \nabla u = -1, \qquad t = u + v, \tag{3.21}$$

where $u$ and $v$ are the distances travelled from the two boundaries and their sum is the thickness.

The cone-ray method pairs each endocardial point with the epicardial point reached along its surface normal, but guards against a single badly oriented ray. Following the shape diameter function of Shapira

et al. [41], $K$ rays are cast within a cone of half-angle $30^\circ$ around the normal, and the median of the valid intersection distances is kept,

$$t_i^{\text{cone}} = \text{median}_{k=1}^{K}\, d_{ik}, \qquad K = 7. \tag{3.22}$$

The estimate keeps an approximately transmural interpretation, that is, it measures across the wall rather than along it, while being more robust than a single normal ray. Figure 3.8 contrasts the four pairing rules on one wall cross-section.

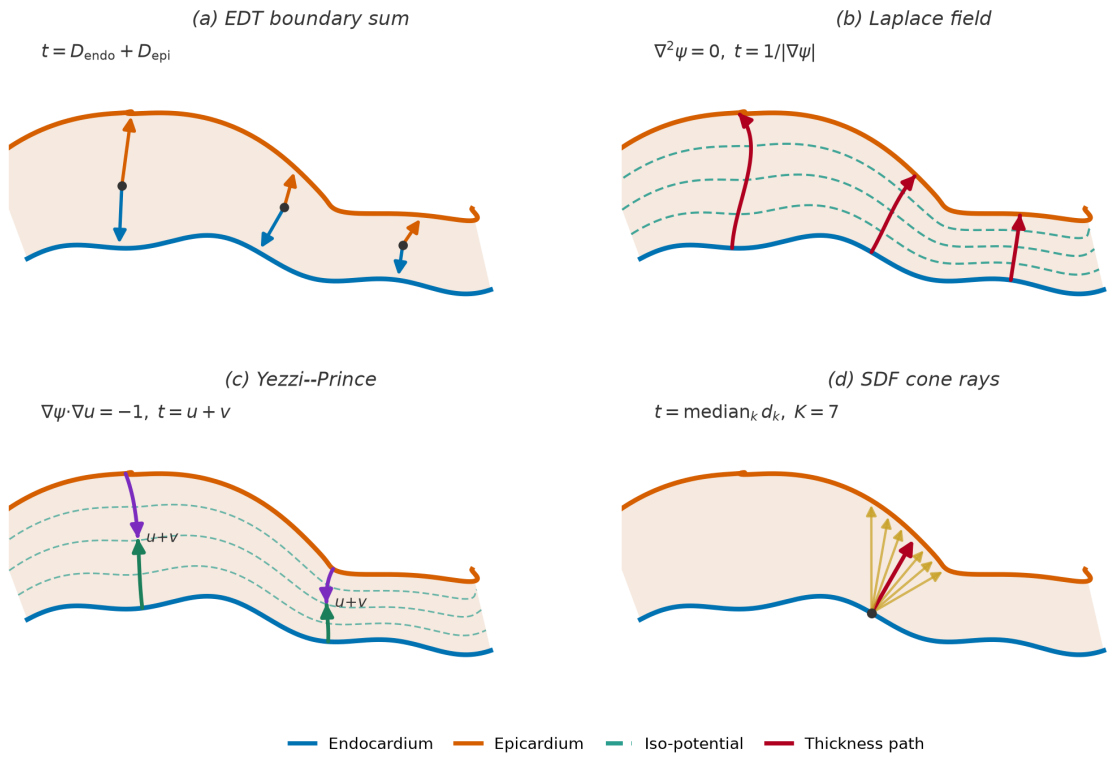



**Figure 3.8.** Conceptual comparison of the four wall-thickness estimators on a single myocardial cross-section, with the endocardium in blue and the epicardium in orange. (a) The EDT boundary sum adds the distances from an interior point to each boundary. (b) The Laplace field method follows the curved paths of a smooth potential that rises from 0 on the endocardium to 1 on the epicardium; the paths cross the level lines of the potential (dashed) at right angles, and thickness is estimated from the gradient magnitude. (c) The Yezzi–Prince method propagates two travel distances $u$ and $v$ from opposite boundaries along that same curved field and sums them. (d) The SDF cone-ray method casts a fan of rays within a cone around the endocardial normal and takes the median intersection distance (highlighted), making it robust to any single ill-oriented ray.

What is specific to this work is not the estimators but how they are used: all four are run on the same reconstructed surfaces under the same conditions, and their agreement with one another, rather than with a reference that does not exist, decides which one carries the regional analysis.

### 3.5.2. Local three-dimensional wall thickness

Before any regional summary, each estimator produces a thickness value at every endocardial vertex of the repaired meshes of Section 3.4.2, that is, a continuous map over the whole reconstructed wall rather than a handful of measurements on selected slices. This map is the primary output of the measurement stage. It shows the gradual change from the thin apex to the thicker base and preserves local thinning or

thickening that averaging within an AHA segment would hide, and a fixed colour scale makes the maps comparable across patients and phases (Figure 3.9).

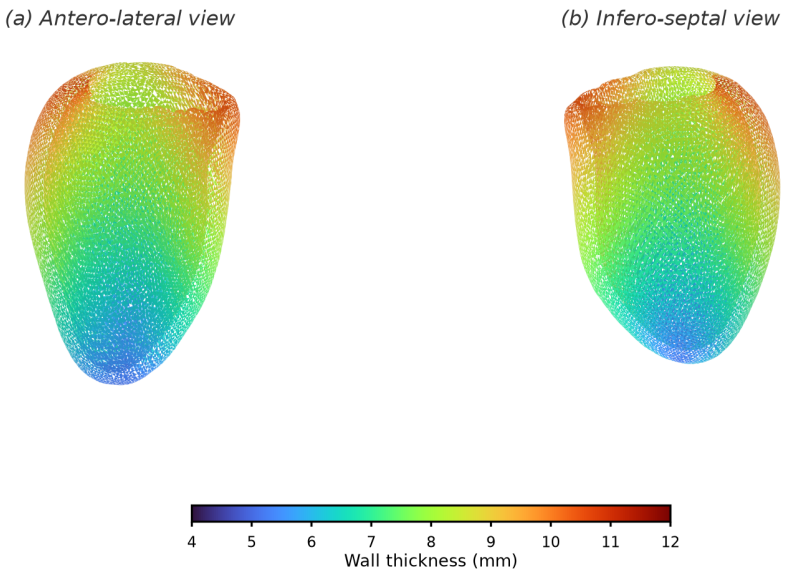



**Figure 3.9.** Local three-dimensional wall-thickness map on the reconstructed LV endocardial surface, shown from two opposing viewpoints. Each vertex is coloured by its local wall thickness on a fixed millimetre scale, so that the continuous spatial variation from the thin apex to the thicker base is visible directly on the geometry rather than only as regional averages.

### 3.5.3. Regional AHA-17 mapping

For clinical reporting, the local map of the selected estimator is summarised in the AHA 17-segment model [6]. The model was defined for two-dimensional imaging. The ventricle is divided along its long axis into basal, mid-cavity, and apical thirds; each third is read on a representative SAX slice and divided into angular sectors around the centre of the cavity, six in the basal and mid-cavity thirds and four in the apical third, numbered from the anterior wall with the septal segments facing the right ventricle; and the tip of the apex forms the seventeenth segment. In routine use, the thickness of each segment is then read on that slice at points chosen by the observer.

Transferring this definition to a reconstructed surface requires only that every vertex receive the two coordinates the definition relies on, a position along the long axis and an angle around it; the segments then become patches of the three-dimensional surface rather than readings on single slices. The long axis is taken as the direction of the SAX stack. The base and apex are the first and last slices that contain myocardium, the base being the end whose cavity cross-section is larger, and every endocardial vertex is given a normalised position between 0 at the base and 1 at the apex. Vertices in the first third of this range form the basal ring, those in the second third the mid-cavity ring, those between two thirds and 0.85 the apical ring, and those beyond 0.85 the apical cap, which is segment 17. The angle of each vertex is measured in the SAX plane around the centre of the endocardial vertices, with its zero fixed by the right-ventricular label of the segmentation: the direction from the LV centre to the centre of the right ventricle marks the septum, the wall shared by the two ventricles, and the anterior reference lies

a quarter turn from it. The basal and mid-cavity rings are then cut into six sectors of $60^\circ$ and the apical ring into four sectors of $90^\circ$, giving segments 1–6, 7–12, and 13–16 (Figure 3.10). A quality-assurance step visualises the detected base, apex, and septal direction before the segments are formed. Unrolling the rings onto concentric circles gives the familiar bullseye, with the basal ring outermost and the apical cap at the centre; each cell of the bullseye now summarises all vertices of one surface patch. Within each segment the vertex values are combined with robust summaries, the mean and percentiles, so that isolated outliers do not dominate the regional value. The aggregation does not replace the per-vertex maps; it restates the same measurements in a form clinicians are used to.

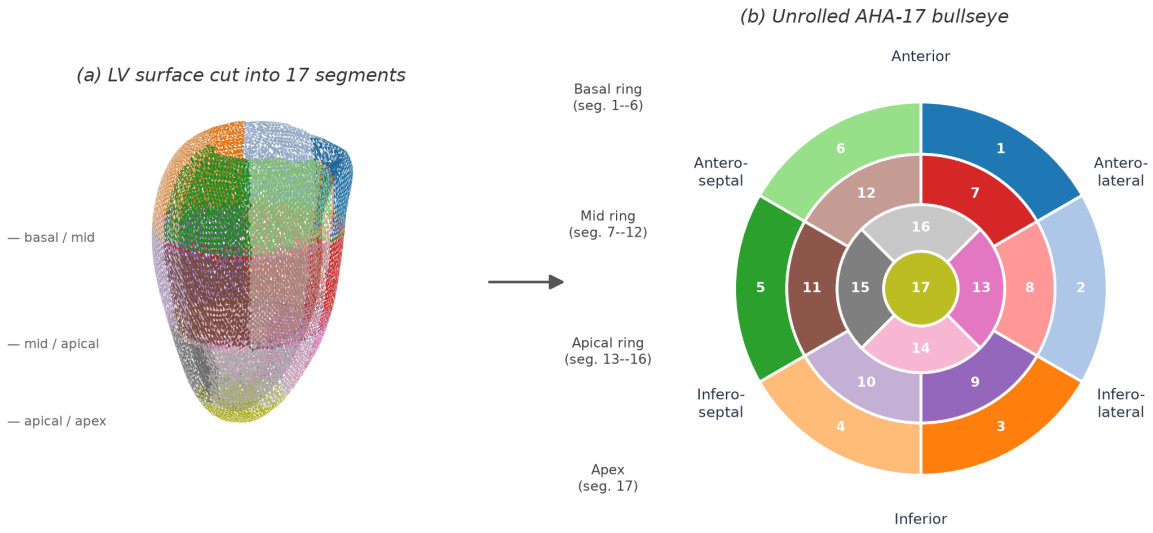



**Figure 3.10.** AHA-17 partitioning of the reconstructed LV surface and corresponding bullseye representation.

This completes the pipeline of Figure 3.1. Its outputs, two reconstructed surfaces per case, four local thickness fields on them, and AHA-17 summaries of the selected field, are the material of Chapter 4. There, agreement with the observed rings and with the RBF and SSM fits addresses the first research question; the behaviour of the coupled surfaces across ED and ES addresses the second; and the agreement among thickness estimators and the regional comparisons address the third.

CHAPTER 4

# Results and Evaluation

This chapter presents the experimental evaluation of the approach described in Chapter 3.

## 4.1. Experimental Setup

The evaluation uses the final model, obtained by pre-training on the synthetic SSM meshes and then fine-tuning on the real end-diastolic (ED) and end-systolic (ES) data, as described in Chapter 3. The datasets and patient-level splits are reported in Section 3.2.3. The general reconstruction and wall-thickness evaluations use the held-out test patients from both ACDC and M&Ms-2. The pathology-stratified regional analysis uses a separate ACDC subset of $30$ test patients, comprising $20$ normal (NOR) and $10$ hypertrophic cardiomyopathy (HCM) cases. The HCM group is included so that the model can be examined on abnormal as well as normal LV geometry. No independent 3D ground truth is available, so the following results characterise contour fidelity and inter-method consistency rather than reconstruction accuracy. Three questions guide the evaluation: how faithfully the model follows the observed LV contours and how far it agrees with alternative reconstructions of them, whether it represents the ED–ES change while maintaining a valid wall, and how closely the resulting thickness measurements agree with those obtained from an independent estimator on the same input.

### 4.1.1. Metrics

Reconstruction geometry is characterised with complementary surface and volume quantities, following the metric framework of Taha and Hanbury [42] and the conventions used for cardiac segmentation [7] and for implicit-surface learning [12]. These quantities are used in two roles: between a reconstruction and the contour rings that produced it, and between two reconstructions of the same rings. In the first role they measure fidelity to the observations; in the second they measure disagreement between methods, and are symmetric, so they belong to neither method as an error.

The *Chamfer distance* is the average distance between two surfaces, taken in both directions. It is reported for the endocardium and the epicardium in millimetres. The *contour distance* is the corresponding one-sided quantity from each observed ring point to the reconstructed surface, summarised by its mean and its 95th percentile.

The *Dice similarity coefficient* (DSC) measures the solid rather than the surface. After voxelisation it is twice the shared volume divided by the sum of the two volumes. It is computed for the endocardial cavity and for the myocardium and lies between zero and one. Enclosed *volumes* are reported in millilitres for each method separately. A volume ratio is deliberately not reported: it would require designating one reconstruction as correct, which the available data do not support.

The *watertight rate* is the fraction of reconstructions that form a closed surface without holes. It is stated in the text as a precondition rather than as a score, because a mesh that is not closed has no volume for the DSC to use.

The 95th-percentile Hausdorff distance (HD95) describes the tail of the distribution and is reported with the Chamfer distance. The average symmetric surface distance is omitted because it tracked the Chamfer distance to within $0.06$ mm throughout. The intersection over union is omitted because it is a direct transformation of the DSC; the F-score depends on one selected tolerance, and normal consistency measures orientation rather than positional agreement.

Wall thickness is summarised by its mean, its standard deviation, and its 5th and 95th percentiles across the myocardial surface, all in millimetres. The percentiles describe the thin apical and thick basal extremes without being distorted by isolated outliers. When the two cardiac phases are compared, the absolute thickening $\Delta_{\mathrm{ES}-\mathrm{ED}}$ and the relative thickening as a percentage of the end-diastolic value are also reported. These are the standard descriptors of regional contractile function.

The four wall-thickness methods are compared with one another on the reconstructed geometry, using the agreement framework of Bland and Altman [43] instead of a comparison of mean values. The reported quantities are the *Pearson correlation coefficient* $r$, the *mean absolute error* (MAE) and *root-mean-square error* (RMSE) in millimetres, and the Bland–Altman *bias* with its *95% limits of agreement* (LoA) (bias $\pm 1.96\,\sigma$). Absolute consistency is summarised by the two-way *intraclass correlation coefficient* (ICC), read against the thresholds of Koo and Li [44]. These are reported together because correlation alone does not show agreement: two methods can rise and fall together and still differ by a constant amount, which only the bias reveals.

### 4.1.2. Comparators

The reconstruction is compared with two alternative reconstructions of the same input contour rings: a radial-basis-function implicit fit, which interpolates the rings through a smooth field, and a fit of the public statistical shape model (SSM), which regularises them towards a population prior. The two were chosen because they impose different and well-understood priors, so the spread between the three results indicates how much the reconstruction prior itself determines the geometry. Both consume the same rings as the model and pass through the same mesh repair and metric pipeline, as described in Section 3.4.2, so their numbers are directly comparable. They are evaluated only at end-diastole: the public SSM is distributed with end-diastolic modes only.

The RBF fit is a reconstruction comparator rather than ground truth. It belongs to the same family as the surrogate targets used for fine-tuning, so close agreement with the model is partly expected and is not independent confirmation. The SSM fit is independent of the training targets, but it carries a population prior that the cohort partly falls outside. Wall thickness is compared between the model, the RBF fit, and the SSM fit with the four algorithms selected in Section 3.5.1: the Laplace field method, the Yezzi–Prince method, the SDF cone-ray method, and the Euclidean Distance Transform (EDT) boundary sum. Because the SAX stacks are sparse relative to their in-plane resolution, all three reconstruction methods must infer the through-plane geometry, and their absolute millimetre values should be read with care.

## 4.2. Training

The final model is produced in two stages. A first stage pre-trains the signed-distance decoder for roughly $200$ epochs on the synthetic SSM meshes. This stage provides a geometric prior, and its checkpoint initializes the second stage. The reported model is then fine-tuned on the real end-diastolic and end-systolic data described in Chapter 3. The available training history records the fine-tuning stage; no numerical claim is made about the pre-training loss because its per-epoch log was not retained.

Fine-tuning runs for $776$ epochs with the AdamW optimiser (learning rate $2 \times 10^{-5}$, weight decay $5 \times 10^{-4}$) under a cosine-annealing schedule with warm restarts, mixed-precision arithmetic, and an effective batch size of $32$. The composite objective is the five-term loss of Equation (3.17), combining surface, dense SDF, Eikonal, off-surface, and wall-thickness supervision. Figure 4.1 plots the total loss and its principal components together with two optimisation diagnostics. The total training loss decreases from $10.07$ at the first fine-tuning epoch to $0.91$, and the validation loss settles at $0.70$, with the best checkpoint selected at epoch $695$ before mild over-fitting appears. The surface and Eikonal terms fall by more than an order of magnitude, indicating successful optimisation of the surface-fitting and Eikonal objectives. Consistently, the mean gradient norm $\|\nabla f\|$ stabilises at $0.996$, close to the unit value encouraged by the Eikonal term.

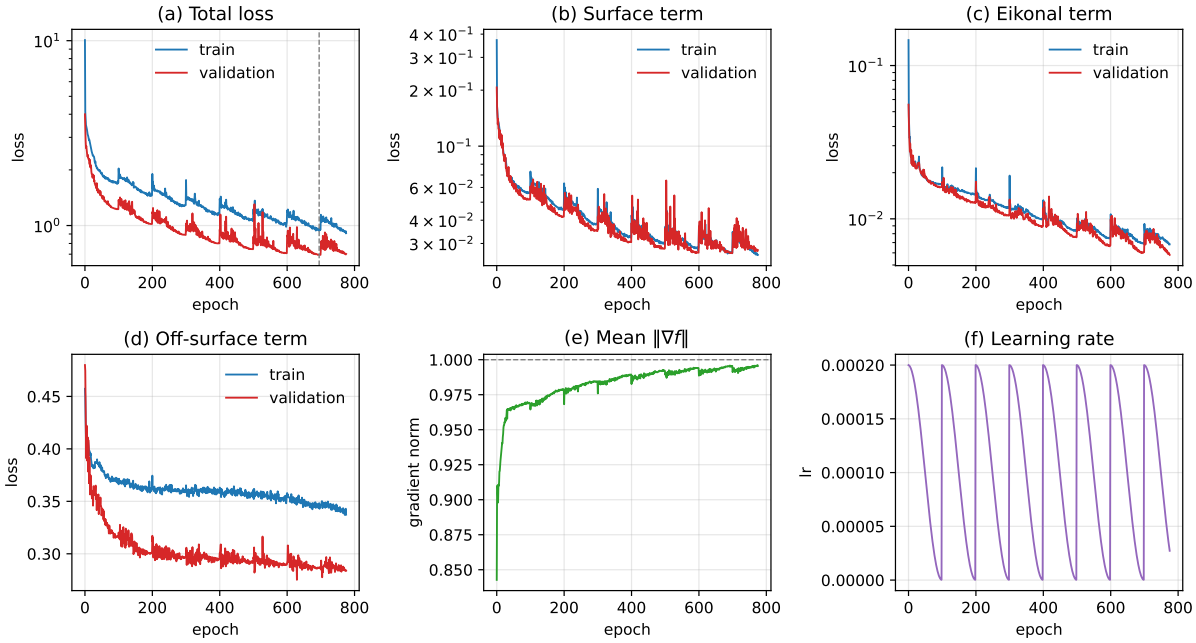



**Figure 4.1.** Training history. Panels (a)–(d) show the total loss and its surface, Eikonal, and off-surface components for the training and validation splits; the dashed vertical line marks the best epoch. Panels (e)–(f) show the mean field gradient norm, whose target is one, and the cosine-annealing learning-rate schedule. The synthetic pre-training stage is not shown because its optimisation history was not retained.

## 4.3. Results

The evaluation covers reconstruction consistency, wall-thickness measurement, and regional analysis with the AHA 17-segment model.

### 4.3.1. Reconstruction Consistency

Contour fidelity measures how closely each surface follows its input rings, while pairwise agreement measures how strongly the reconstruction prior changes the completed geometry. These measures describe consistency rather than accuracy because no independent 3D reference is available. All values below are end-diastolic and cover the held-out ACDC and M&Ms-2 test cohort. Every surface from every method is watertight after the shared repair stage.

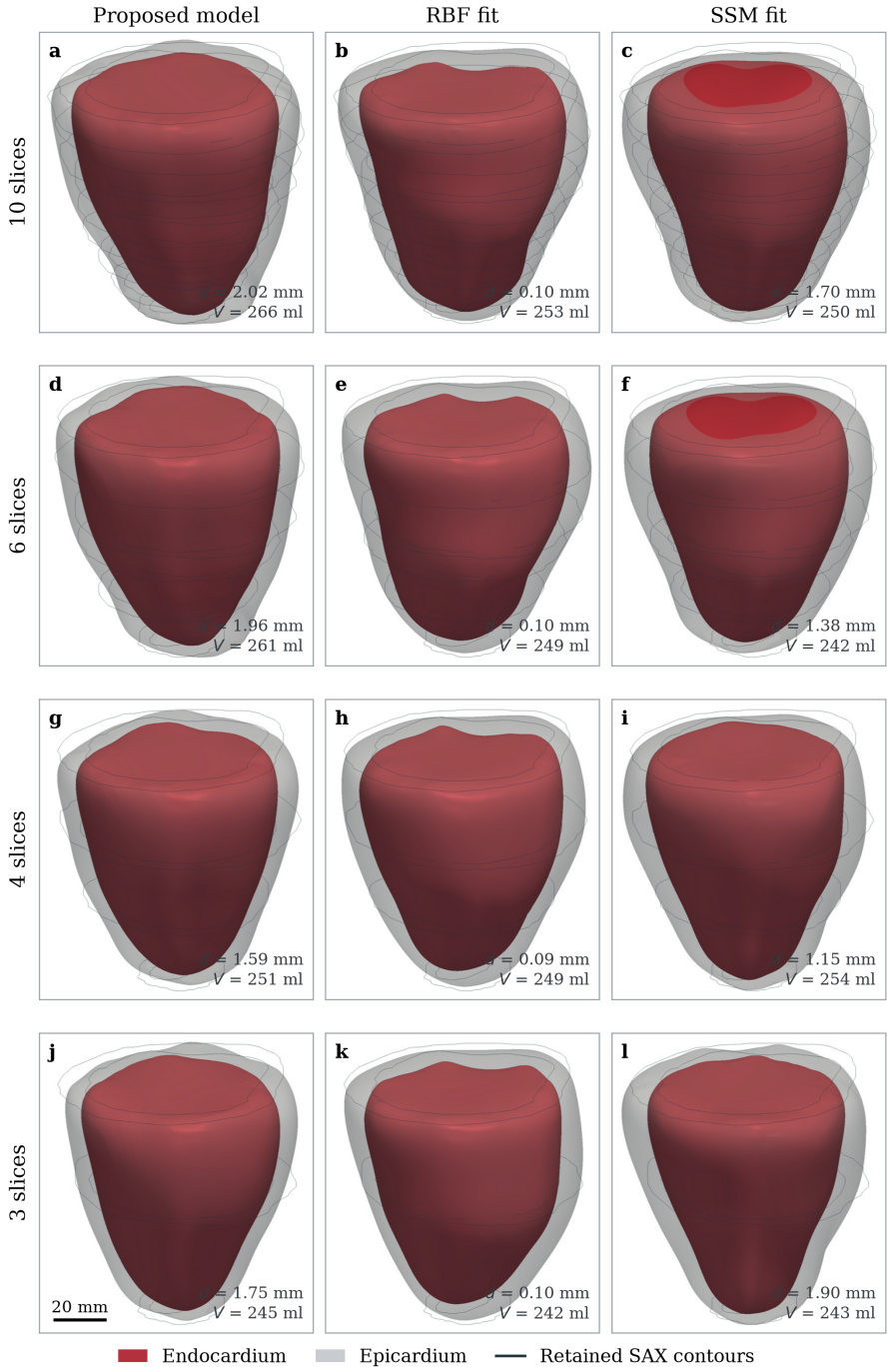



**Figure 4.2.** Left-ventricular surfaces reconstructed from a progressively decimated end-diastolic SAX stack. Columns compare the proposed model, RBF implicit fit, and SSM fit; rows show ten, six, four, and three retained slices. Red and grey surfaces show the endocardium and epicardium, respectively, and dark rings show the input contours. All panels share the same view and scale. The panels illustrate one ACDC case; the corresponding quantitative analysis in Table 4.3 covers the held-out test cohort.

Figure 4.2 provides a qualitative comparison of the three methods and previews the slice-sensitivity analysis.

In the complete ten-slice row, all three methods recover the same overall ventricular shape from identical observations, but they encode different priors. The RBF fit interpolates the retained contours, the SSM fit regularises the surface towards a population shape, and the proposed model combines a learned prior with conditioning on the observed rings. The cohort-level differences between these reconstructions are examined first through contour fidelity and pairwise geometric agreement. The remaining rows of the figure are then analysed separately as a sensitivity test of inference under reduced slice density.

Table 4.1 reports the distance from each input contour ring to the reconstructed surface and the enclosed volumes.

**Table 4.1.** Distance from the input SAX contour rings to the surface reconstructed from them, and the enclosed volumes, at end-diastole over the held-out ACDC and M&Ms-2 test cohort. Values are mean $\pm$ standard deviation of the per-patient statistic; $p_{95}$ is the cohort mean of the per-patient 95th percentile. Bold marks the lowest distance on each surface. The RBF fit interpolates the contour points by construction, so its residual is a boundary condition of the method rather than a reconstruction advantage. Volumes are not ranked because no reference volume is available.

| Method | Surface | Contour distance (mm) | $p_{95}$ (mm) | Volume (ml) |
|---|---|---|---|---|
| Proposed model | Endocardium | $1.50 \pm 0.23$ | $3.56$ | $119.3 \pm 31.3$ |
| | Epicardium | $1.96 \pm 0.28$ | $5.23$ | $255.3 \pm 76.7$ |
| RBF fit | Endocardium | $\mathbf{0.13 \pm 0.01}$ | $\mathbf{0.30}$ | $118.6 \pm 30.2$ |
| | Epicardium | $\mathbf{0.11 \pm 0.01}$ | $\mathbf{0.26}$ | $233.5 \pm 71.4$ |
| SSM fit | Endocardium | $1.82 \pm 0.67$ | $5.51$ | $126.9 \pm 32.2$ |
| | Epicardium | $1.91 \pm 0.76$ | $6.23$ | $215.2 \pm 56.3$ |

The implicit fit reproduces the rings almost exactly, at $0.13$ and $0.11$ mm on the two surfaces. That figure must not be read as an order-of-magnitude advantage. The fit is an interpolant whose defining constraint is that the field vanish at exactly those contour points, so reproducing them is a boundary condition of the method and the measured residual is the solver's, not the reconstruction's. Any interpolating method scores near zero here and any regularising method does not, whatever their relative merits between the slices. The quantity is reported because it verifies that each method honours its input and because it separates interpolating from regularising behaviour, not because it ranks the methods.

Two further limits apply. The rings are separated by several millimetres, so a metric anchored on them says nothing about the intervening volume, which is where the reconstruction problem actually lies. And exact fidelity to the rings is not unambiguously desirable, because the contours carry annotation noise and in-plane discretisation: a surface that follows them to $0.13$ mm follows their errors as well.

The model deviates from the rings by $1.50$ mm on the endocardium and $1.96$ mm on the epicardium, which is of the order of the in-plane pixel size of $1.37$ to $1.88$ mm, and is therefore consistent with a surface that smooths annotation noise rather than one that misses the observations. The SSM fit deviates by a similar amount on average, at $1.82$ and $1.91$ mm, but far less predictably: its per-patient endocardial figure spans $0.92$ to $3.44$ mm against $1.20$ to $2.14$ mm for the model.

Within the $30$-patient ACDC pathology subset, the diagnostic split explains that spread. The SSM fit is close to the observations on the normal cases, at $1.42$ mm, and drifts on the hypertrophic ones, at $2.63$ mm, which is the expected behaviour of a population prior asked to represent anatomy outside its training variation. The model is comparatively stable across the two groups, at $1.45$ and $1.60$ mm, and the implicit fit is insensitive to pathology by construction.

Enclosed volumes tell a complementary story. The three methods agree closely on the cavity, at $119$, $119$ and $127$ ml, but diverge on the myocardium, at $136$, $115$ and $88$ ml. The wall, not the cavity, is where the reconstruction priors disagree, which matters directly for the thickness measurements reported in Section 4.3.3.

Table 4.2 compares the three reconstructions with one another. Because none of them is a reference, each entry is a symmetric disagreement between two methods rather than an error of either. The average symmetric surface distance is omitted because it differed from the Chamfer distance by less than $0.06$ mm for every pair.

**Table 4.2.** Pairwise agreement between the three reconstructions of the same end-diastolic contour rings, over the held-out ACDC and M&Ms-2 test cohort. Values are mean $\pm$ standard deviation of the per-patient statistic. Every entry is a symmetric disagreement between two methods, not an error, because no method is a reference. Distances are in millimetres; Dice is dimensionless. Bold marks the closest pair for each metric and surface.

| Pair | Metric | Endocardium/cavity | Epicardium/myocardium |
|---|---|---|---|
| Model vs. RBF fit | Chamfer (mm) | **1.57 ± 0.18** | **1.90 ± 0.24** |
| | HD95 (mm) | **4.24 ± 0.86** | **6.07 ± 0.94** |
| | Dice | **0.91 ± 0.02** | **0.77 ± 0.05** |
| Model vs. SSM fit | Chamfer (mm) | $2.08 \pm 0.51$ | $2.59 \pm 0.73$ |
| | HD95 (mm) | $5.88 \pm 1.91$ | $6.72 \pm 2.32$ |
| | Dice | $0.88 \pm 0.03$ | $0.69 \pm 0.04$ |
| RBF fit vs. SSM fit | Chamfer (mm) | $2.01 \pm 0.67$ | $2.05 \pm 0.64$ |
| | HD95 (mm) | $5.88 \pm 2.95$ | $6.62 \pm 2.67$ |
| | Dice | $0.89 \pm 0.04$ | $0.76 \pm 0.06$ |

The model and the implicit fit are the closest pair, differing by $1.57$ mm on the endocardium and $1.90$ mm on the epicardium, with a myocardial Dice of $0.77$. The two pairs involving the SSM are further apart, at $2.08$ and $2.01$ mm on the endocardium. The ordering is the same for every surface metric, which places the learned reconstruction nearer to a smooth interpolant of the observations than to a population prior.

The disagreement between any two methods, between $1.6$ and $2.1$ mm on the endocardium, is comparable to the in-plane pixel size and far smaller than the $10$ mm slice spacing, so all three priors agree about the ventricle to within roughly one pixel where the data constrain them. Cavity agreement is uniformly high, at Dice $0.88$ to $0.91$, while myocardial agreement is markedly lower, at $0.69$ to $0.77$, and its patient-level range is wide: myocardial Dice falls to $0.58$ for the worst SSM comparison. The reconstructions agree about the cavity but disagree about the wall, which is the quantity this thesis ultimately measures. This spread sets a floor on the uncertainty of any wall-thickness value reported

later in this chapter, and it cannot be reduced by choosing one of these methods as though it were correct.

The higher cavity agreement than myocardial agreement reflects the available data. The contour rings constrain the cavity shape at each acquired slice. The two surfaces must still be inferred between slices, where small changes can alter myocardial volume and thickness without greatly changing cavity overlap. Lower myocardial Dice therefore shows that wall geometry is sensitive to the reconstruction prior. It does not show that one method is anatomically better.

### 4.3.2. Slice Sensitivity

Having established the differences between reconstruction methods on the complete stacks, this section examines how their inference changes when through-plane information is removed, over the held-out test cohort. Each patient's complete end-diastolic stack is reduced to six, four and three slices while always retaining the most basal and most apical slice, so the axial support is fixed and only the through-plane sampling density changes. Because the acquired stacks differ in length, a patient contributes to a given level only when its complete stack has strictly more slices than that level. The remaining rows of Figure 4.2 illustrate the effect on one case. All three methods continue to produce closed, plausible surfaces with only three slices, so none fails outright under this reduction.

Table 4.3 quantifies the change by comparing each decimated reconstruction with the surface produced by the same method from that patient's complete stack. These within-method deviations measure self-consistency rather than anatomical error and isolate sensitivity to missing slices from the between-method differences reported above.

**Table 4.3.** Deviation from each method's complete-stack reconstruction at end-diastole over the held-out test cohort. Distances are in millimetres and Dice is dimensionless; $\Delta V_{\text{myo}}$ is the change in myocardial volume relative to the complete-stack reconstruction. Each patient is compared with its own complete stack and contributes to a level only when its stack has strictly more slices than that level. All meshes are watertight, and metrics use the untrimmed and unsmoothed surfaces. Bold marks the smallest deviation among the three methods at each slice level: the lowest distance, the highest Dice, and the smallest absolute $\Delta V_{\text{myo}}$.

| | | Chamfer (mm) | | Dice | | |
|---|---|---|---|---|---|---|
| **Method** | **Slices** | **Endo.** | **Epi.** | **Cavity** | **Myoc.** | $\Delta V_{\text{myo}}$ **(%)** |
| Proposed model | 6 | **0.79** | 0.87 | **0.971** | **0.891** | −0.4 |
| | 4 | 1.27 | 1.48 | 0.953 | 0.819 | −2.1 |
| | 3 | 1.83 | 2.17 | 0.934 | 0.782 | −3.8 |
| RBF fit | 6 | 0.83 | **0.83** | 0.968 | 0.887 | **0.0** |
| | 4 | **1.15** | **0.88** | **0.959** | **0.858** | **−1.8** |
| | 3 | **1.45** | **1.32** | **0.947** | **0.805** | −6.2 |
| SSM fit | 6 | 0.89 | 0.95 | 0.968 | 0.883 | −1.8 |
| | 4 | 1.46 | 1.39 | 0.942 | 0.812 | −3.8 |
| | 3 | 1.94 | 1.92 | 0.921 | 0.775 | **+2.6** |

### 4.3.3. Wall-Thickness Measurement

The four selected methods are applied to the proposed reconstruction, the RBF fit, and the SSM fit of every patient at end-diastole. Table 4.4 reports, for each estimator and reconstruction, the cohort average of the per-patient mean, standard deviation, and 5th and 95th percentiles.

**Table 4.4.** Wall-thickness statistics at end-diastole over the held-out ACDC and M&Ms-2 test cohort. Each entry is the cohort average of the corresponding per-patient statistic, in millimetres, before any calibration.

| Method | Geometry | Mean (mm) | SD (mm) | $p_5$ (mm) | $p_{95}$ (mm) |
|---|---|---|---|---|---|
| Laplace field | Model | 9.18 | 2.14 | 4.75 | 12.20 |
| | RBF fit | 8.36 | 2.21 | 5.03 | 12.29 |
| | SSM fit | 6.48 | 1.38 | 3.87 | 8.43 |
| Yezzi–Prince | Model | 10.01 | 1.64 | 7.41 | 12.74 |
| | RBF fit | 8.86 | 2.15 | 5.57 | 12.71 |
| | SSM fit | 7.10 | 1.23 | 4.75 | 8.88 |
| SDF cone rays | Model | 12.47 | 4.87 | 8.01 | 20.68 |
| | RBF fit | 9.38 | 2.49 | 5.63 | 13.80 |
| | SSM fit | 7.11 | 1.50 | 4.12 | 9.13 |
| EDT boundary sum | Model | 9.55 | 1.66 | 6.62 | 12.17 |
| | RBF fit | 8.51 | 2.00 | 5.42 | 12.05 |
| | SSM fit | 7.02 | 1.22 | 4.65 | 8.77 |

The three non-ray methods agree on the magnitude of the wall within each reconstruction. Their means range from $9.18$ to $10.01$ mm on the proposed reconstruction, from $8.36$ to $8.86$ mm on the RBF fit, and from $6.48$ to $7.10$ mm on the SSM fit. Holding the estimator fixed, the reconstruction prior changes the absolute thickness scale. The SDF cone-ray method gives the largest value on the proposed reconstruction and the RBF fit, at $12.47$ and $9.38$ mm, respectively, but not on the SSM fit.

Dispersion separates the methods the same way. The per-patient standard deviation is $1.64$ mm for Yezzi–Prince, $1.66$ mm for the EDT boundary sum and $2.14$ mm for the Laplace field on the proposed reconstruction, but $4.87$ mm for the cone-ray estimator. Its 5th-to-95th-percentile span, $8.01$ to $20.68$ mm, is about twice as wide as that of any other estimator on that geometry, making the ray-based estimator both the highest and the noisiest of the four on the model output.

Table 4.5 compares patient-level mean thickness on the proposed reconstruction with the two fitting baselines.

**Table 4.5.** Patient-level agreement between wall thickness measured on the proposed reconstruction and on the two fitting baselines at end-diastole. Bias is model minus comparator; bias and RMSE are in millimetres, and the Pearson correlation $r$ is dimensionless. Bold marks the closest agreement in each column: the smallest bias and RMSE and the highest $r$.

| | Model vs. RBF fit | | | Model vs. SSM fit | | |
|---|---|---|---|---|---|---|
| **Method** | **Bias (mm)** | **RMSE (mm)** | $r$ | **Bias (mm)** | **RMSE (mm)** | $r$ |
| Laplace field | $+$**0.82** | **0.86** | **0.99** | $+2.70$ | 3.13 | 0.72 |
| Yezzi–Prince | $+1.16$ | 1.23 | **0.99** | $+2.91$ | 3.49 | **0.77** |
| SDF cone rays | $+3.10$ | 3.81 | 0.95 | $+5.36$ | 6.64 | 0.67 |
| EDT boundary sum | $+1.03$ | 1.10 | **0.99** | $+$**2.53** | **3.04** | **0.77** |

Applying the same estimator to all three geometries isolates the effect of the reconstruction prior. Relative to the model, the RBF fit is thinner by $0.82$ mm for the Laplace field, $1.16$ mm for Yezzi–Prince, $3.09$ mm for SDF cone rays, and $1.03$ mm for the EDT boundary sum. The SSM fit is thinner by $2.70$, $2.91$, $5.36$, and $2.53$ mm, respectively. Per-patient means remain strongly correlated between the model and RBF fit ($r = 0.99$, $0.99$, $0.95$, and $0.99$), whereas agreement with the SSM fit is more moderate ($r = 0.72$, $0.77$, $0.67$, and $0.77$). The RBF fit largely preserves between-patient variation while shifting the absolute scale; the stronger SSM prior changes both the scale and, to a lesser extent, the patient ordering.

Table 4.6 compares the three other methods with the Laplace field point by point over the myocardial surface.

**Table 4.6.** Agreement of each wall-thickness method with the Laplace field, over the myocardial surface points of the held-out ACDC and M&Ms-2 test cohort. MAE, RMSE, bias and the 95% LoA are in millimetres; the Pearson correlation $r$ and the intraclass correlation coefficient (ICC) are dimensionless. The Laplace field is the common comparator, not a ground truth. Bold marks the closest agreement in each column: the highest $r$ and ICC and the lowest MAE, RMSE, and absolute bias.

| **Method** | $r$ | **MAE (mm)** | **RMSE (mm)** | **Bias (95% LoA) (mm)** | **ICC** |
|---|---|---|---|---|---|
| Yezzi–Prince | **0.80** | 0.89 | 2.14 | $+0.83$ ($\pm 3.86$) | **0.77** |
| SDF cone rays | 0.44 | 3.38 | 7.29 | $+3.31$ ($\pm 12.74$) | 0.28 |
| EDT boundary sum | 0.78 | **0.81** | **2.07** | $+$**0.37** ($\pm 3.99$) | **0.77** |

The two remaining non-ray methods track it closely. The EDT boundary sum differs from it by $0.81$ mm on average with a bias of $+0.37$ mm, and Yezzi–Prince by $0.89$ mm with a bias of $+0.83$ mm; both reach ICC $= 0.77$. The cone-ray method differs by $3.38$ mm on average, with a bias of $+3.31$ mm, limits of agreement of $\pm 12.74$ mm and ICC $= 0.28$. Read against the thresholds of Koo and Li [44], the first two show moderate to good agreement with the Laplace field and the third poor agreement. The correlations make the same point in a different way: all three exceed $r = 0.44$, but $r$ alone would not have revealed a bias of more than three millimetres.

The Laplace field is used for the regional analysis of Section 4.3.4. This choice is methodological rather than numerical. Although introduced for cortical thickness rather than cardiac imaging [36], it provides a transferable construction for paired tissue boundaries. It measures thickness along the streamlines of a harmonic field spanning the two surfaces, so the path stays inside the wall wherever

the wall curves. A straight ray does not: it leaves the myocardium early or late in curved regions, which is what the cone-ray numbers above show. The results support that choice because the Laplace field agrees more closely with the two non-ray methods than with the cone-ray estimator.

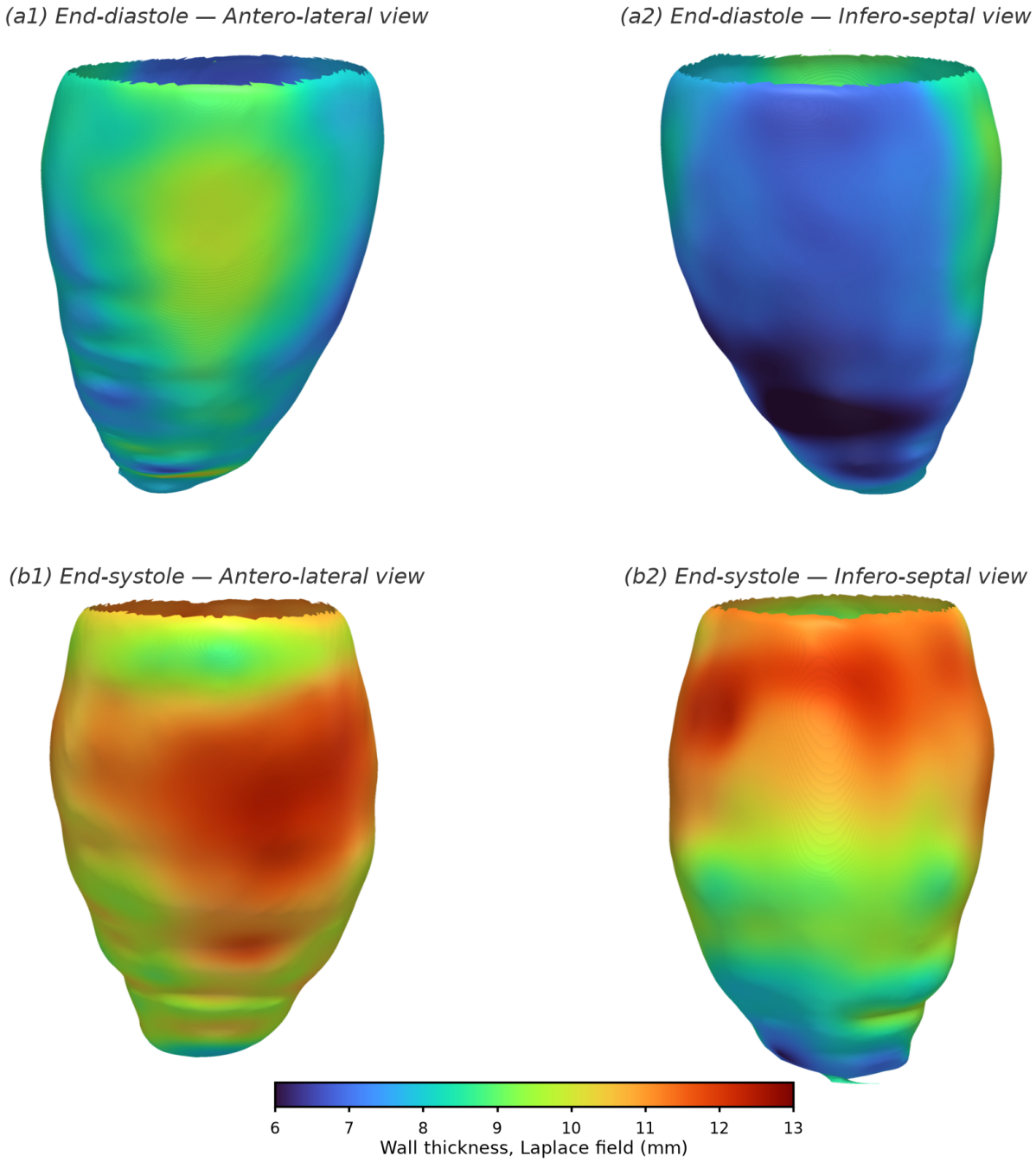



**Figure 4.3.** Calibrated analytic offset map on the reconstructed endocardial surface of one representative case from the ACDC cohort, at end-diastole (top) and end-systole (bottom), each seen from two opposing viewpoints. Every vertex is coloured by the decoder wall offset evaluated at that point, calibrated per phase to the distance-transform reference of the input segmentation, on a scale shared by all four panels. The absolute colour level is therefore inherited from that reference. The surface is trimmed at the most basal input contour, so the base ends in an open rim and only the observed part of the wall is shown; above that plane the field is pure extrapolation.

The decoder returns a continuous wall-offset field rather than a geometric thickness measurement. Figure 4.3 visualises this field for the representative case at both cardiac phases, in the same form as

the methodology illustration of Figure 3.9 but computed on the model output instead of on the SSM mean mesh. The decoder offset is defined only up to the normalisation applied to the input contours, so its absolute level is calibrated per phase to the mean of the distance-transform reference measured on the input segmentation, exactly as the cohort pipeline does; the calibration factor is close to two for both phases. The figure is thus a calibrated analytic offset map, not a wall-thickness measurement or an independently predicted absolute magnitude. The residual circumferential banding visible on the surfaces marks the through-plane spacing of the input SAX slices, where the field is interpolated between observations.

Figure 4.3 visualises the calibrated decoder offset emitted directly by the model, whereas Table 4.8 reports Laplace-derived wall thickness measured between the post-processed endocardial and epicardial meshes. The offset map is a qualitative model-output visualisation; it does not provide the values used for the AHA-17 analysis.

Because the public SSM provides no ES modes, the fitting-baseline comparison is limited to ED and phase-dependent wall thickening is reported on the reconstructed geometry in the regional analysis below.

### 4.3.4. Regional Analysis over the AHA 17-Segment Model

To localise the measurements anatomically, the reconstructed surface is partitioned into the seventeen segments of the American Heart Association (AHA) model, assigning each myocardial vertex to a basal, mid-cavity, apical, or apical-cap segment from its normalised long-axis height and its circumferential angle relative to the right-ventricular insertion.

Figure 4.4 shows the partition applied to the reconstructed geometry of the representative case together with a bullseye of the Laplace-derived wall thickness.

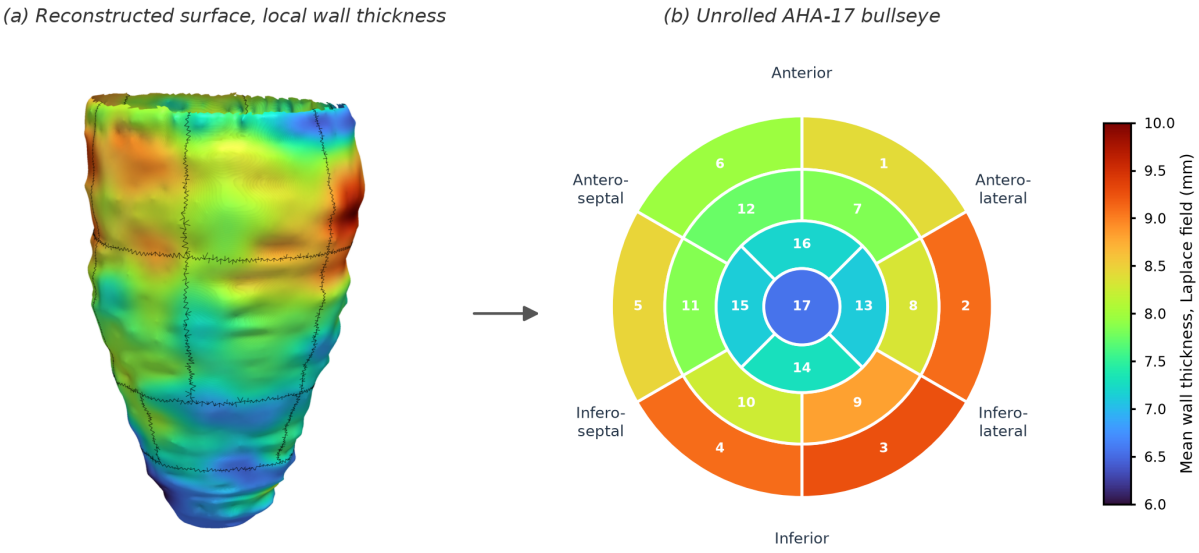



**Figure 4.4.** AHA 17-segment analysis of the reconstructed left-ventricular geometry at end-diastole for the representative case. Panel (a) shows the reconstructed surface partitioned into the seventeen segments, each coloured by its mean wall thickness measured with the Laplace field method and separated from its neighbours by a black boundary line; panel (b) unrolls the same segment values onto the standard bullseye, where basal segments form the outer ring (1–6), mid-cavity segments the middle ring (7–12), apical segments the inner ring (13–16), and the apical cap the centre (17). The two panels share the colour scale, so the bullseye is a flattening of the geometry beside it rather than an independent summary.

All regional values are measured with the Laplace field, for the reasons given in Section 4.3.3. Table 4.7 reports the regional pattern over the $30$-patient ACDC pathology subset and its change between phases.

**Table 4.7.** Regional AHA 17-segment wall thickness on the reconstructed geometry over the $30$-patient ACDC pathology subset.

| # | Segment | ED (mm) | ES (mm) | $\Delta$ (%) |
|---|---|---|---|---|
| 1 | Basal anterior | 9.99 | 13.71 | $+37.1$ |
| 2 | Basal anteroseptal | 10.15 | 13.40 | $+32.1$ |
| 3 | Basal inferoseptal | 9.89 | 13.43 | $+35.7$ |
| 4 | Basal inferior | 9.09 | 13.81 | $+51.9$ |
| 5 | Basal inferolateral | 8.93 | 14.27 | $+59.8$ |
| 6 | Basal anterolateral | 9.53 | 13.87 | $+45.5$ |
| 7 | Mid anterior | 9.19 | 13.79 | $+50.0$ |
| 8 | Mid anteroseptal | 10.37 | 14.68 | $+41.6$ |
| 9 | Mid inferoseptal | 10.19 | 15.05 | $+47.6$ |
| 10 | Mid inferior | 9.25 | 14.41 | $+55.8$ |
| 11 | Mid inferolateral | 9.09 | 14.96 | $+64.6$ |
| 12 | Mid anterolateral | 8.99 | 14.17 | $+57.6$ |
| 13 | Apical anterior | 8.50 | 13.63 | $+60.3$ |
| 14 | Apical septal | 8.77 | 13.86 | $+58.1$ |
| 15 | Apical inferior | 8.44 | 13.29 | $+57.5$ |
| 16 | Apical lateral | 8.43 | 13.78 | $+63.6$ |
| 17 | Apex | 7.76 | 11.69 | $+50.7$ |
| **Overall** | | **9.21** | **13.87** | $+$**50.6** |

*Note.* Values are means for the $30$ ACDC test patients used in the pathology analysis, measured with the Laplace field. ED and ES denote end-diastole and end-systole, respectively; $\Delta$ is the relative thickening from ED to ES.

Over the ACDC pathology subset the reconstructed wall averages $9.21$ mm at ED and $13.87$ mm at ES, corresponding to $50.6\%$ relative thickening. At ED the septal mid-cavity segments are the thickest, while the apical cap is the thinnest at $7.76$ mm. Every segment thickens between phases; the smallest relative change is $32.1\%$ in the basal anteroseptal segment and the largest is $64.6\%$ in the mid inferolateral segment. These phase values describe the model output and are not an accuracy comparison, because neither fitting baseline provides a matched regional ES evaluation.

Table 4.8 separates the normal and hypertrophic patients, showing whether the reconstruction represents differences associated with abnormal ventricular geometry or simply reproduces an average shape.

**Table 4.8.** Regional AHA 17-segment wall thickness on the reconstructed geometry for the normal and hypertrophic groups.

| # | Segment | Normal (NOR, $n = 20$) | | | Hypertrophic (HCM, $n = 10$) | | |
|---|---|---|---|---|---|---|---|
| | | ED | ES | $\Delta$ | ED | ES | $\Delta$ |
| | | (mm) | | (%) | (mm) | | (%) |
| 1 | Basal anterior | 8.72 | 12.25 | $+40.4$ | 12.54 | 16.62 | $+32.6$ |
| 2 | Basal anteroseptal | 8.81 | 12.10 | $+37.4$ | 12.83 | 16.02 | $+24.9$ |
| 3 | Basal inferoseptal | 8.68 | 11.96 | $+37.8$ | 12.32 | 16.35 | $+32.8$ |
| 4 | Basal inferior | 8.16 | 12.53 | $+53.6$ | 10.96 | 16.37 | $+49.4$ |
| 5 | Basal inferolateral | 7.94 | 12.81 | $+61.3$ | 10.91 | 17.19 | $+57.5$ |
| 6 | Basal anterolateral | 8.38 | 12.12 | $+44.6$ | 11.83 | 17.37 | $+46.9$ |
| 7 | Mid anterior | 7.94 | 12.10 | $+52.5$ | 11.70 | 17.16 | $+46.6$ |
| 8 | Mid anteroseptal | 8.55 | 12.98 | $+51.7$ | 13.99 | 18.10 | $+29.4$ |
| 9 | Mid inferoseptal | 8.57 | 12.98 | $+51.4$ | 13.43 | 19.17 | $+42.8$ |
| 10 | Mid inferior | 7.93 | 12.46 | $+57.1$ | 11.89 | 18.32 | $+54.1$ |
| 11 | Mid inferolateral | 7.96 | 13.02 | $+63.5$ | 11.35 | 18.84 | $+66.0$ |
| 12 | Mid anterolateral | 7.92 | 12.39 | $+56.4$ | 11.13 | 17.73 | $+59.3$ |
| 13 | Apical anterior | 7.28 | 11.75 | $+61.4$ | 10.96 | 17.40 | $+58.7$ |
| 14 | Apical septal | 7.49 | 12.00 | $+60.2$ | 11.33 | 17.59 | $+55.3$ |
| 15 | Apical inferior | 7.28 | 11.72 | $+61.1$ | 10.76 | 16.43 | $+52.6$ |
| 16 | Apical lateral | 7.33 | 12.03 | $+64.0$ | 10.61 | 17.28 | $+62.9$ |
| 17 | Apex | 6.80 | 10.32 | $+51.8$ | 9.67 | 14.71 | $+52.1$ |
| **Overall** | | **7.99** | **12.21** | $+$**52.9** | **11.68** | **17.23** | $+$**47.5** |

*Note.* Values are group means measured with the Laplace field. ED and ES denote end-diastole and end-systole, respectively; $\Delta$ is the relative thickening from ED to ES. These measurements summarise the model output reconstructed from sparse SAX contours and are not an independent anatomical reference.

The regional pattern is internally coherent in both groups. In the normal cases the end-diastolic thickness varies little around the base and the mid-cavity, from $7.92$ to $8.81$ mm, and tapers towards the apex, where the apical cap at $6.80$ mm is the thinnest region. Every segment thickens from end-diastole to end-systole, and the thickening is graded along the long axis: the apical segments thicken by $52$% to $64$%, the mid-cavity segments by $51$% to $64$% and the basal segments by $37$% to $61$%. Over the normal group the mean thickness rises from $7.99$ mm to $12.21$ mm, a thickening of $52.9$%.

The hypertrophic group is thicker in every segment. The largest differences are in the mid-cavity anteroseptal and inferoseptal segments, where the HCM group reaches $13.99$ and $13.43$ mm compared with $8.55$ and $8.57$ mm in the normal group. Table 4.9 evaluates this separation at the patient level rather than treating the 17 segments as independent observations.

**Table 4.9.** Patient-level contrasts between the normal and HCM groups. Values are mean ± standard deviation. Differences are HCM minus NOR with a 95% percentile-bootstrap confidence interval from 10 000 patient-level resamples. Positive Hedges' $g$ indicates a larger value in HCM.

| Endpoint | NOR | HCM | Difference (95% CI) | Hedges' $g$ |
|---|---|---|---|---|
| Mean ED AHA-17 thickness (mm) | $7.99 \pm 0.88$ | $11.66 \pm 1.66$ | $+3.67\ [2.54,\ 4.67]$ | 3.01 |
| Maximum ED segment (mm) | $9.07 \pm 0.98$ | $14.44 \pm 1.94$ | $+5.37\ [4.11,\ 6.56]$ | 3.82 |
| Systolic thickening (percentage points) | $53.2 \pm 11.6$ | $47.9 \pm 11.5$ | $-5.3\ [-13.9,\ 3.0]$ | $-0.45$ |

The two end-diastolic confidence intervals exclude zero and their effect sizes are large, supporting descriptive separation in absolute thickness. In contrast, the interval for relative systolic thickening includes zero; this cohort does not establish a clear group difference in proportional thickening. Despite the maximum-segment contrast, none of the normal cases and only three of the ten HCM cases exceed the $15$ mm reference line. The corresponding clinical criterion applies to maximal end-diastolic LV wall thickness in adults when the hypertrophy is not explained by loading conditions or another disease [45]. The endpoint reported here is instead the maximum of the reconstructed AHA-17 segment means and must not be interpreted as a diagnostic result.

The difference between groups also varies by region. The largest end-diastolic differences occur in the mid-cavity septal segments. The apical cap is the thinnest region in both groups. These are descriptive patterns in the reconstructed surfaces, not evidence of segment-level diagnostic performance. The regional values from one patient are related, and no independent three-dimensional reference is available.

Figure 4.5 shows the patient-level distributions of absolute ED thickness and relative systolic thickening.

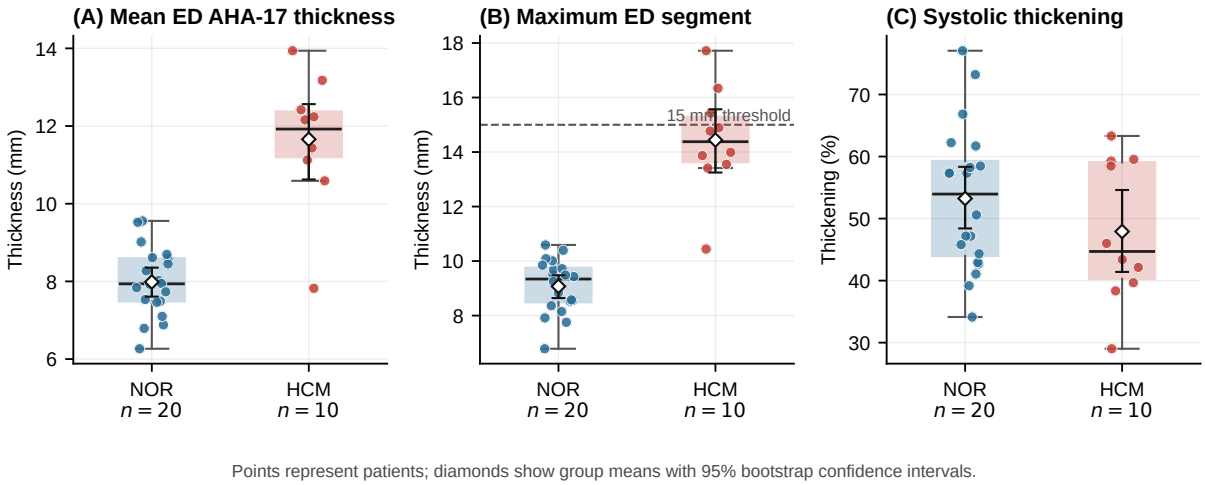



**Figure 4.5.** Patient-level AHA-17 wall-thickness outcomes for the normal and HCM groups. The dashed line shows the 15 mm clinical reference for maximal end-diastolic LV wall thickness [45]; the plotted endpoint is the maximum reconstructed AHA-17 segment mean and is not diagnostically equivalent to that criterion.

The groups overlap more for relative thickening than for absolute thickness.

Each patient contributes seventeen segment values, so the segment observations are not independent of one another. Segments from the same heart share the same reconstruction, the same input

contours and the same anatomy. For this reason the segment values are used as a descriptive regional summary, and the group comparisons above are made at the patient level, where each patient counts once. The segment-level numbers should not be treated as if they came from independent observations.

## 4.4. Discussion

The results address the three research questions stated in Section 1.3, although the available evidence answers some components more directly than others.

For **RQ1**, on whether a continuous, watertight surface can be reconstructed from sparse contours and how it compares with conventional alternatives, the end-diastolic evaluation shows that the model follows the observed contour rings with mean distances of $1.50$ mm on the endocardium and $1.96$ mm on the epicardium (Table 4.1). After the shared repair stage, all evaluated meshes are watertight; closure is therefore a property of the complete reconstruction pipeline rather than of raw surface extraction alone. Pairwise Chamfer distances are $1.57$ and $1.90$ mm against the RBF fit and $2.08$ and $2.59$ mm against the SSM fit for the endocardium and epicardium, respectively (Table 4.2). In the absence of independent 3D ground truth, these measurements establish contour fidelity and inter-method consistency, not anatomical accuracy.

The slice-decimation analysis further shows that all three repaired methods remain closed with three retained slices, and that their myocardial agreement with the corresponding complete-stack surface degrades to a similar degree: Dice falls to $0.782$ for the model, $0.805$ for the RBF fit and $0.775$ for the SSM fit, while myocardial volume changes by $-3.8$%, $-6.2$% and $+2.6$%, respectively (Table 4.3). The model shows the largest epicardial displacement at that level, at $2.17$ mm, but not the largest volume change, so no method is uniformly more robust to slice density than the others. Because each method is compared with its own complete-stack result, these figures demonstrate sensitivity to slice density rather than anatomical error.

For **RQ2**, on whether a single model can represent multiple physiological states while preserving a plausible separation between the two surfaces, the model represents separate endocardial and epicardial surfaces at ED and ES, and the Laplace-derived regional mean thickness increases in all seventeen segments, from $9.21$ mm at ED to $13.87$ mm at ES overall (Table 4.7). The current evaluation does not, however, report a pointwise minimum wall separation, a surface-intersection or collapse check, or the ED–ES change in cavity volume. The evidence therefore supports phase-dependent representation and wall thickening, but does not establish positive separation at every evaluated point or phase-specific reconstruction accuracy. The absence of matched ES modes for the SSM further limits the phase comparison.

For **RQ3**, on how closely measurements derived from the reconstructed geometry agree with those from reference clinical geometry and how that agreement depends on the measurement method, applying the same estimator to the proposed and RBF reconstructions yields correlations of $r = 0.95$ to $0.99$ in patient-level mean thickness, but positive biases of $0.82$ to $3.10$ mm show that strong correlation does not imply agreement in absolute scale (Table 4.5). This is a global comparison between geometries. Separately, pointwise comparisons on the proposed geometry show that Yezzi–Prince and the

EDT boundary sum agree substantially more closely with the Laplace field than does the SDF cone-ray estimator (Table 4.6). Together, these results demonstrate dependence on both reconstruction geometry and measurement method, but local and regional agreement with the RBF-derived geometry remains unreported.

The pathology-stratified analysis provides additional descriptive evidence. Relative to the normal group, the HCM group has a $3.67$ mm higher patient-level mean ED thickness and a $5.37$ mm higher maximum ED segment, with both confidence intervals excluding zero (Table 4.9). The difference in relative systolic thickening is less conclusive at $-5.3$ percentage points because its confidence interval includes zero. These contrasts describe separation within the reconstructed outputs and do not constitute diagnostic validation.

Taken together, the complete pipeline converts sparse contours into repaired, watertight paired surfaces that remain close to the observations and exhibit method-dependent agreement with conventional ED reconstructions. Patient-level mean thickness preserves relative variation most strongly against the RBF fit, and the reconstructed geometry supports descriptive regional analysis. These findings do not establish anatomical accuracy, pointwise wall validity across phases, or local and regional thickness agreement, because no independent 3D ground truth is available.

## 4.5. Threats to Validity

The most important limitation concerns the reference. No public dataset provides independently acquired 3D ground-truth surfaces for these scans, so the reconstruction is evaluated through fidelity to the observed contour rings and agreement with alternative reconstructions of those rings. The pairwise Chamfer distances measure disagreement between reconstruction priors, not distance from true anatomy. The same caveat applies to the wall-thickness comparison: the input is a stack of contours separated by several millimetres, so no method used in this chapter carries measured through-plane information, and the way that gap is inferred shifts the measured thickness. Absolute millimetre values should not be over-interpreted; the comparisons that the data support are relative ones between pipelines evaluated on identical input. The wall-thickness study is an inter-method and reproducibility exercise, not a clinical validation. Without expert manual wall-thickness measurements, this work can claim an objective and locally resolved computational framework, but not the replacement of clinical measurement protocols.

Several further factors bound the external validity of the results, and the whole evaluation should be read as preliminary. The general evaluation uses held-out patients from ACDC and M&Ms-2, whereas the pathology-stratified regional analysis is limited to $30$ ACDC test patients: $20$ normal and $10$ with hypertrophic cardiomyopathy. This subset is far too small to support claims of clinical or pathological generalisation, and the hypertrophic subgroup in particular is only large enough for a directional observation. The regional analysis carries a further dependency: each patient contributes seventeen segments, so the segment values are repeated measures within a patient rather than independent samples, and they are reported as a descriptive summary for that reason. The relative thickening figures deserve particular caution, because dividing a systolic by a diastolic thickness amplifies small differences in where the wall is sampled. The regional summaries also depend on the AHA orientation, while all mesh-based

measurements depend on the inference grid. A finer grid or an alternative anatomical frame could shift the values. Only the decoder-offset map in Figure 4.3 is calibrated to the segmentation reference; the quantitative tables report geometric estimators before calibration. The reconstruction additionally inherits the quality of the input segmentation, and the training data span more than one collection, so some domain shift between acquisition sources is expected. Finally, the synthetic epicardium used during pre-training is a geometric construction rather than a measured surface, which the fine-tuning stage on real data is designed to correct but cannot fully rule out as a residual prior. A larger cohort, and ideally a dataset with genuine 3D ground truth, would be needed to confirm the results reported here.

The comparison between reconstructions is bounded in the same way. The implicit fit and the SSM fit were chosen because their priors differ from each other and from the learned one, so the spread between the three bounds the influence of the prior; it does not identify which is closest to the anatomy. The implicit fit is moreover of the same family as the surrogate targets used to fine-tune the model, so its agreement with the model is not an independent check. Contour fidelity cannot settle the question either, because it evaluates only the observed rings and not the geometry between them. The slice-decimation analysis in Table 4.3 compares each method with its own complete-stack reconstruction, so it measures self-consistency rather than anatomical error. That criterion may favour a stable prior that responds weakly to the observations as well as a method that generalises well from them. Because the acquired stacks vary in length, the three decimation levels are also supported by different subsets of patients, and the deepest level rests on the longest stacks alone. The comparison does not cover spline lofting, Poisson reconstruction, or other learned models, and the SSM comparison is limited to end-diastole because the public model provides no end-systolic modes. No controlled ablation removes synthetic pre-training, local conditioning, phase conditioning, or positive surface coupling. The experiment characterises the model against two conventional alternatives, but it cannot attribute its behaviour to a specific component or establish that the full architecture is necessary.

CHAPTER 5

# Conclusions

This chapter closes the thesis. It summarises what was built and why, answers the three research questions of Section 1.3 on the evidence of Chapter 4, states the contributions, delimits the claims, and sets out the work that should follow.

## 5.1. Summary of the Work

The thesis started from a geometric limitation of short-axis (SAX) cardiac MRI. Each slice observes the left ventricle (LV) finely, at an in-plane pixel size below $2$ mm, but the slices lie several millimetres apart, so the three-dimensional wall on which clinical measurement depends is never observed directly and has to be inferred. We treated that inference as the learning of a continuous signed-distance representation. A phase-conditioned implicit model was pre-trained on synthetic shapes sampled from a statistical shape model (SSM) and fine-tuned on end-diastolic (ED) and end-systolic (ES) surfaces fitted to ACDC and M&Ms-2 segmentations. The model reads the sparse contour rings through a global PointNet pathway and a local feature grid, predicts the endocardial signed-distance field, and defines the epicardial field as that field minus a strictly positive, directly supervised offset, so the two surfaces cannot cross. Its output was compared with a radial-basis-function (RBF) implicit fit and an SSM fit of the same rings, and myocardial wall thickness was measured on all three geometries with four local estimators, localised on the AHA 17-segment model, and contrasted between normal and hypertrophic patients.

## 5.2. Answers to the Research Questions

**RQ1** asked to what extent a continuous, watertight surface can be reconstructed from sparse contours, and how such a reconstruction compares with conventional implicit and shape-model alternatives. On the held-out ACDC and M&Ms-2 test patients, the model reproduces its input rings to within $1.50$ mm on the endocardium and $1.96$ mm on the epicardium, which is the order of the in-plane pixel size, and every surface it produces is watertight after the shared repair stage. Its geometry lies closer to the RBF interpolant (Chamfer distance $1.57$ and $1.90$ mm) than to the SSM fit ($2.08$ and $2.59$ mm), and it responds to pathology more evenly than the population prior: between the normal and hypertrophic subsets its mean contour distance moves from $1.45$ to $1.60$ mm, whereas that of the SSM fit moves from $1.42$ to $2.63$ mm. The comparison also shows where the three priors diverge. They agree on the cavity, with a Dice coefficient of $0.88$ to $0.91$ and enclosed volumes of $119$, $119$ and $127$ ml, but not on the wall, where the Dice coefficient falls to $0.69$ to $0.77$ and the myocardial volume ranges from $88$ to $136$ ml. Removing slices down to three per stack degrades all three methods by a similar amount without breaking any of them. A closed, contour-faithful surface can therefore be recovered from sparse SAX rings, but the wall

completed between the slices remains a property of the prior rather than of the data, and the spread between methods is the honest measure of that uncertainty.

**RQ2** asked whether a single model can represent multiple physiological states while preserving a physically plausible separation between the inner and outer surfaces. One phase-conditioned network reconstructs both ED and ES geometry, and because the epicardial field is the endocardial field minus a positive offset, the ordering of the two surfaces is fixed by the architecture rather than checked after the fact. On the reconstructed surfaces, the Laplace-derived wall thickness rises from $9.21$ mm at ED to $13.87$ mm at ES over the ACDC pathology subset, and all seventeen AHA segments thicken, with the smallest relative change at the base and the largest in the mid-cavity and apical segments. The evaluation did not, however, measure a pointwise minimum separation on the extracted meshes, test for intersections after repair, or report the ED–ES change in cavity volume, and the public SSM offers no ES modes against which the systolic geometry could be compared. RQ2 therefore receives a qualified answer: the model represents both phases with a wall that thickens consistently between them, but positive separation is established by construction of the fields and by regional means, not verified at every point of the output.

**RQ3** asked how closely measurements derived from the reconstructed geometry agree with those from reference clinical geometry, and how that agreement depends on the measurement method. Both factors matter, and they act in different ways. With the estimator held fixed, the reconstruction prior shifts the absolute scale of the wall: patient-level mean thickness on the model correlates with that on the RBF fit at $r = 0.99$ for the three non-ray estimators, yet is $0.82$ to $1.16$ mm thicker, and against the SSM fit the correlation falls to $0.72$ to $0.77$ with a bias of $2.53$ to $2.91$ mm. With the geometry held fixed, the Yezzi–Prince method and the EDT boundary sum agree with the Laplace field to within a bias below $1$ mm and an intraclass correlation of $0.77$, whereas the SDF cone-ray estimator is biased by $+3.31$ mm with limits of agreement of $\pm 12.74$ mm and an intraclass correlation of $0.28$, because a straight ray leaves a curved wall where a streamline does not. Strong correlation between geometries thus does not imply agreement in millimetres; the reconstruction prior and the measurement path are separate sources of offset of comparable size, and path-following estimators are the appropriate choice on a curved wall. Because the RBF fit is a smooth-surface comparator rather than an anatomical reference, these figures bound the reproducibility of the measurement, not its accuracy.

The pathology-stratified analysis adds that the reconstructed geometry retains patient-specific variation instead of collapsing towards an average shape. The hypertrophic group is thicker in every AHA segment; its patient-level mean ED thickness exceeds that of the normal group by $3.67$ mm (95% CI $[2.54,\ 4.67]$) and its thickest segment by $5.37$ mm ($[4.11,\ 6.56]$), both with large effect sizes, whereas relative systolic thickening does not separate the groups. These are descriptive separations within the model output on ten hypertrophic patients, not a diagnostic result.

## 5.3. Contributions

The thesis makes three contributions. The first is a data pipeline that brings synthetic SSM meshes and clinical ED and ES segmentations from two public datasets into one sample format, with contour rings,

surface samples, dense signed-distance supervision and wall-offset targets, so that a model can be pre-trained on population geometry and fine-tuned on patient data without a change of representation. The second is the phase-conditioned implicit model, whose encoder pools endocardial and epicardial evidence separately before merging it and complements the global code with a local feature grid read at every query point, and whose decoder couples the epicardial field to the endocardial field through a directly supervised positive offset. The third is an evaluation protocol for sparse-contour reconstruction in the absence of 3D ground truth: fidelity to the observed rings, symmetric agreement between differently motivated reconstructions, self-consistency under slice decimation, and a wall-thickness comparison that separates the effect of the geometry from the effect of the estimator before the values are localised on the AHA model and contrasted between patient groups. The protocol does not depend on this particular network, and its central finding, that reconstruction priors agree about the cavity and disagree about the wall, concerns the problem rather than the model.

## 5.4. Limitations

The answers above hold within the limits set out in Section 4.5. No independent three-dimensional reference exists for these scans, so every geometric figure in the thesis measures fidelity to the rings or agreement between methods, never distance from the true anatomy, and every thickness figure describes reproducibility rather than validated measurement. Because the RBF comparator belongs to the same family as the surrogate surfaces used for fine-tuning, its close agreement with the model is partly expected. Twenty normal and ten hypertrophic ACDC patients support a directional observation about pathology and no more, and the seventeen segment values of each patient are repeated measures within one heart rather than independent samples. The SSM comparison is confined to end-diastole, no ablation isolates the contribution of synthetic pre-training, local conditioning, phase conditioning or the positive-offset coupling, and the pre-training loss history was not retained. What the evaluation offers is a careful characterisation of a working system, not a validation of it.

## 5.5. Future Work

The limitations fix the order of the next steps. The first need is an anatomical reference. Higher-resolution volumetric acquisitions, or expert three-dimensional surface and wall-thickness annotations on a larger external cohort, would turn the agreement figures reported here into accuracy figures and would show which of the three priors lies closest to the true wall. The second is a wider and fully matched comparison, adding spline lofting, Poisson reconstruction and learned alternatives under the same patients, physical spacing, repair procedure and metrics, with pointwise error maps that localise disagreement at the apex, the base and the most curved parts of the wall. The third is the controlled ablation that this thesis did not run: removing synthetic pre-training, local conditioning, phase conditioning and the positive-offset coupling in turn would show which components the behaviour depends on, and adding pointwise separation and intersection checks on the extracted meshes would complete the answer to RQ2. Replacing the binary phase label with a continuous position in the cardiac cycle would then allow the representation to be tested for coherence in time as well as continuity in space.

In sum, this thesis shows that a phase-conditioned implicit representation can turn a sparse SAX stack into closed, paired LV surfaces that honour the observed contours, thicken consistently between diastole and systole, and support locally resolved wall-thickness analysis. It also shows, through the disagreement between equally plausible reconstructions of the same rings, how much of the myocardial wall is still inferred rather than observed, and therefore how carefully any thickness derived from it must be read.

# Bibliography


[1] W. Grossman et al., “Wall thickness and diastolic properties of the left ventricle,” *Circulation*, vol. 49, no. 1, pp. 129–135, 1974. DOI: 10.1161/01.CIR.49.1.129

[2] A. de Marvao, T. J. W. Dawes, W. Shi, et al., “Population-based studies of myocardial hypertrophy: High resolution cardiovascular magnetic resonance atlases improve statistical power,” *Journal of Cardiovascular Magnetic Resonance*, vol. 16, p. 16, 2014. DOI: 10.1186/1532-429X-16-16

[3] P. Medrano-Gracia, B. R. Cowan, B. Ambale-Venkatesh, et al., “Left ventricular shape variation in asymptomatic populations: The multi-ethnic study of atherosclerosis,” *Journal of Cardiovascular Magnetic Resonance*, vol. 16, p. 56, 2014. DOI: 10.1186/s12968-014-0056-2

[4] W. Bai, W. Shi, A. de Marvao, et al., “A bi-ventricular cardiac atlas built from 1000+ high resolution MR images of healthy subjects and an analysis of shape and motion,” *Medical Image Analysis*, vol. 26, no. 1, pp. 133–145, 2015. DOI: 10.1016/j.media.2015.08.009

[5] W. Bai, M. Sinclair, G. Tarroni, et al., “Automated cardiovascular magnetic resonance image analysis with fully convolutional networks,” *Journal of Cardiovascular Magnetic Resonance*, vol. 20, p. 65, 2018. DOI: 10.1186/s12968-018-0471-x

[6] M. D. Cerqueira, N. J. Weissman, V. Dilsizian, et al., “Standardized myocardial segmentation and nomenclature for tomographic imaging of the heart: A statement for healthcare professionals from the cardiac imaging committee of the council on clinical cardiology of the American Heart Association,” *Circulation*, vol. 105, no. 4, pp. 539–542, 2002. DOI: 10.1161/hc0402.102975

[7] O. Bernard, A. Lalande, C. Zotti, et al., “Deep learning techniques for automatic MRI cardiac multi-structures segmentation and diagnosis: Is the problem solved?” *IEEE Transactions on Medical Imaging*, vol. 37, no. 11, pp. 2514–2525, 2018. DOI: 10.1109/TMI.2018.2837502

[8] C. Petitjean and J.-N. Dacher, “A review of segmentation methods in short axis cardiac MR images,” *Medical Image Analysis*, vol. 15, no. 2, pp. 169–184, 2011. DOI: 10.1016/j.media.2010.12.004

[9] C. Brás, S. Ben Haddou, T. P. Kuipers, et al., “Fighting MRI anisotropy: Learning multiple cardiac shapes from a single implicit neural representation,” in *Medical Imaging 2026: Image Processing*, (Vancouver, Canada), Y. Gan and J. Mitra, Eds., vol. 13925, SPIE, 2026, 139250R, 139250R-1–139250R-9. DOI: 10.1117/12.3086312

[10] T. Heimann and H.-P. Meinzer, “Statistical shape models for 3D medical image segmentation: A review,” *Medical Image Analysis*, vol. 13, no. 4, pp. 543–563, 2009. DOI: 10.1016/j.media.2009.05.004

[11] T. F. Cootes, C. J. Taylor, D. H. Cooper, and J. Graham, “Active shape models – their training and application,” *Computer Vision and Image Understanding*, vol. 61, no. 1, pp. 38–59, 1995. DOI: 10.1006/cviu.1995.1004

[12] L. Mescheder, M. Oechsle, M. Niemeyer, S. Nowozin, and A. Geiger, “Occupancy networks: Learning 3D reconstruction in function space,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 4460–4470. DOI: 10.1109/CVPR.2019.00459

[13] J. J. Park, P. Florence, J. Straub, R. A. Newcombe, and S. Lovegrove, “DeepSDF: Learning continuous signed distance functions for shape representation,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 165–174. DOI: 10.1109/CVPR.2019.00025

[14] W. E. Lorensen and H. E. Cline, “Marching cubes: A high resolution 3D surface construction algorithm,” in *Proceedings of the 14th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH)*, ACM, 1987, pp. 163–169. DOI: 10.1145/37402.37422

[15] A. Gropp, L. Yariv, N. Haim, M. Atzmon, and Y. Lipman, “Implicit geometric regularization for learning shapes,” in *Proceedings of the 37th International Conference on Machine Learning*, PMLR, 2020, pp. 3789–3799.

[16] M. Tancik, P. P. Srinivasan, B. Mildenhall, et al., “Fourier features let networks learn high frequency functions in low dimensional domains,” in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 7537–7547. arXiv: 2006.10739.

[17] O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional networks for biomedical image segmentation,” in *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*, Springer, 2015, pp. 234–241. DOI: 10.1007/978-3-319-24574-4_28

[18] F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, “nnU-Net: A self-configuring method for deep learning-based biomedical image segmentation,” *Nature Methods*, vol. 18, no. 2, pp. 203–211, 2021. DOI: 10.1038/s41592-020-01008-z

[19] C. Martín-Isla, V. M. Campello, C. Izquierdo, K. Kushibar, C. Sendra-Balcells, P. Gkontra, et al., “Deep learning segmentation of the right ventricle in cardiac MRI: The M&Ms challenge,” *IEEE Journal of Biomedical and Health Informatics*, vol. 27, no. 7, pp. 3302–3313, 2023. DOI: 10.1109/JBHI.2023.3267857

[20] C. Sudlow, J. Gallacher, N. Allen, et al., “UK Biobank: An open access resource for identifying the causes of a wide range of complex diseases of middle and old age,” *PLOS Medicine*, vol. 12, no. 3, e1001779, 2015. DOI: 10.1371/journal.pmed.1001779

[21] A. F. Frangi, D. Rueckert, J. A. Schnabel, and W. J. Niessen, “Automatic construction of multiple-object three-dimensional statistical shape models: Application to cardiac modeling,” *IEEE Transactions on Medical Imaging*, vol. 21, no. 9, pp. 1151–1166, 2002. DOI: 10.1109/TMI.2002.804426

[22] A. Suinesiaputra et al., “Statistical shape modeling of the left ventricle: Myocardial infarct classification challenge,” *IEEE Journal of Biomedical and Health Informatics*, vol. 22, no. 2, pp. 503–515, 2018. DOI: 10.1109/JBHI.2017.2652449

[23] S. S. Khalafvand et al., “Assessment of human left ventricle flow using statistical shape modelling and computational fluid dynamics,” *Journal of Biomechanics*, vol. 74, pp. 116–125, 2018. DOI: 10.1016/j.jbiomech.2018.04.030

[24] J. C. Carr, R. K. Beatson, J. B. Cherrie, et al., “Reconstruction and representation of 3D objects with radial basis functions,” in *Proceedings of the 28th Annual Conference on Computer Graphics and Interactive Techniques*, ACM, 2001, pp. 67–76. DOI: 10.1145/383259.383266

[25] N. Wang, Y. Zhang, Z. Li, Y. Fu, W. Liu, and Y.-G. Jiang, “Pixel2Mesh: Generating 3D mesh models from single RGB images,” in *Computer Vision – ECCV 2018*, Springer, 2018, pp. 55–71. DOI: 10.1007/978-3-030-01252-6_4

[26] H. Choi, G. Moon, and K. M. Lee, “Pose2Mesh: Graph convolutional network for 3D human pose and mesh recovery from a 2D human pose,” in *Computer Vision – ECCV 2020*, 2020, pp. 769–787. DOI: 10.1007/978-3-030-58571-6_45

[27] A. Suinesiaputra et al., “A collaborative resource to build consensus for automated left ventricular segmentation of cardiac MR images,” *Medical Image Analysis*, vol. 18, no. 1, pp. 50–62, 2014. DOI: 10.1016/j.media.2013.09.001

[28] M. Beetz, A. Banerjee, and V. Grau, “Generating subpopulation-specific biventricular anatomy models using conditional point cloud variational autoencoders,” in *Statistical Atlases and Computational Models of the Heart (STACOM), MICCAI 2021*, Springer, 2022, pp. 75–83. DOI: 10.1007/978-3-030-93722-5_9

[29] M. Muffoletto, U. Hermida, C. A. Mauger, et al., *Neural implicit heart coordinates: 3D cardiac shape reconstruction from sparse segmentations*, 2025. DOI: 10.48550/arXiv.2512.19316 arXiv: 2512.19316.

[30] J. F. Kunz, S. Ruschke, and R. Heckel, "Implicit neural networks with Fourier-feature inputs for free-breathing cardiac MRI reconstruction," *IEEE Transactions on Computational Imaging*, vol. 10, pp. 1280–1289, 2024. DOI: 10.1109/TCI.2024.3452008

[31] L. Quillien, M. Beaumont, D. Mandry, et al., "Sensor-free physiological guidance for free-breathing cardiac cine MRI using implicit neural representation CineJENSE reconstruction," *Physiological Measurement*, vol. 46, no. 9, p. 095 002, 2025. DOI: 10.1088/1361-6579/adfc23

[32] R. Wang, S. Zhong, J. Li, et al., "Real-time 4D MRI reconstruction using DVR-NeMF, a framework for dynamic volumetric reconstruction," *Cell Reports Methods*, vol. 5, no. 12, p. 101 239, 2025. DOI: 10.1016/j.crmeth.2025.101239

[33] X. Tian, L. Chen, Q. Wu, et al., "Unsupervised motion-compensated decomposition for cardiac MRI reconstruction via neural representation," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, 2026, pp. 9529–9537. DOI: 10.1609/aaai.v40i11.37914

[34] C. R. Qi, H. Su, K. Mo, and L. J. Guibas, "PointNet: Deep learning on point sets for 3D classification and segmentation," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017, pp. 652–660. DOI: 10.1109/CVPR.2017.16

[35] T. Huelnhagen et al., "Myocardial effective transverse relaxation time T2* correlates with left ventricular wall thickness: A 7.0 T MRI study," *Magnetic Resonance in Medicine*, vol. 77, no. 6, pp. 2381–2389, 2017. DOI: 10.1002/mrm.26312

[36] S. E. Jones, B. R. Buchbinder, and I. Aharon, "Three-dimensional mapping of cortical thickness using Laplace's equation," *Human Brain Mapping*, vol. 11, no. 1, pp. 12–32, 2000. DOI: 10.1002/1097-0193(200009)11:1<12::AID-HBM20>3.0.CO;2-K

[37] A. J. Yezzi and J. L. Prince, "An Eulerian PDE approach for computing tissue thickness," *IEEE Transactions on Medical Imaging*, vol. 22, no. 10, pp. 1332–1339, 2003. DOI: 10.1109/TMI.2003.817775

[38] S. Peng, M. Niemeyer, L. Mescheder, M. Pollefeys, and A. Geiger, "Convolutional occupancy networks," in *Computer Vision – ECCV 2020*, ser. Lecture Notes in Computer Science, vol. 12348, Cham: Springer, 2020, pp. 523–540. DOI: 10.1007/978-3-030-58580-8_31

[39] C. Shorten and T. M. Khoshgoftaar, "A survey on image data augmentation for deep learning," *Journal of Big Data*, vol. 6, p. 60, 2019. DOI: 10.1186/s40537-019-0197-0

[40] P. Chlap, H. Min, N. Vandenberg, J. Dowling, L. Holloway, and A. Haworth, "A review of medical image data augmentation techniques for deep learning applications," *Journal of Medical Imaging and Radiation Oncology*, vol. 65, no. 5, pp. 545–563, 2021. DOI: 10.1111/1754-9485.13261

[41] L. Shapira, A. Shamir, and D. Cohen-Or, "Consistent mesh partitioning and skeletonisation using the shape diameter function," *The Visual Computer*, vol. 24, no. 4, pp. 249–259, 2008. DOI: 10.1007/s00371-007-0197-5

[42] A. A. Taha and A. Hanbury, "Metrics for evaluating 3D medical image segmentation: Analysis, selection, and tool," *BMC Medical Imaging*, vol. 15, p. 29, 2015. DOI: 10.1186/s12880-015-0068-x

[43] J. M. Bland and D. G. Altman, "Statistical methods for assessing agreement between two methods of clinical measurement," *The Lancet*, vol. 327, no. 8476, pp. 307–310, 1986. DOI: 10.1016/S0140-6736(86)90837-8

[44] T. K. Koo and M. Y. Li, "A guideline of selecting and reporting intraclass correlation coefficients for reliability research," *Journal of Chiropractic Medicine*, vol. 15, no. 2, pp. 155–163, 2016. DOI: 10.1016/j.jcm.2016.02.012

[45] S. R. Ommen, S. Mital, M. A. Burke, et al., “2020 AHA/ACC guideline for the diagnosis and treatment of patients with hypertrophic cardiomyopathy: A report of the american college of cardiology/american heart association joint committee on clinical practice guidelines,” *Circulation*, vol. 142, no. 25, e558–e631, 2020. DOI: 10.1161/CIR.0000000000000937

APPENDIX A

# Supplementary Methodological Details

This appendix summarises the training streams, data partitions, clinical RBF mesh-preparation operations, and model configuration described in Chapter 3.

**Table A.1.** Training data streams and their roles.

| Stream | Phase | Source | Training role |
|---|---|---|---|
| Synthetic SSM | ED | UK Digital Heart Project LV SSM | Pre-training to establish a geometric shape prior |
| Real MRI | ED | ACDC and M&Ms-2 | Fine-tuning on patient-specific diastolic anatomy |
| Real MRI | ES | ACDC and M&Ms-2 | Fine-tuning on systolic anatomy with phase conditioning |

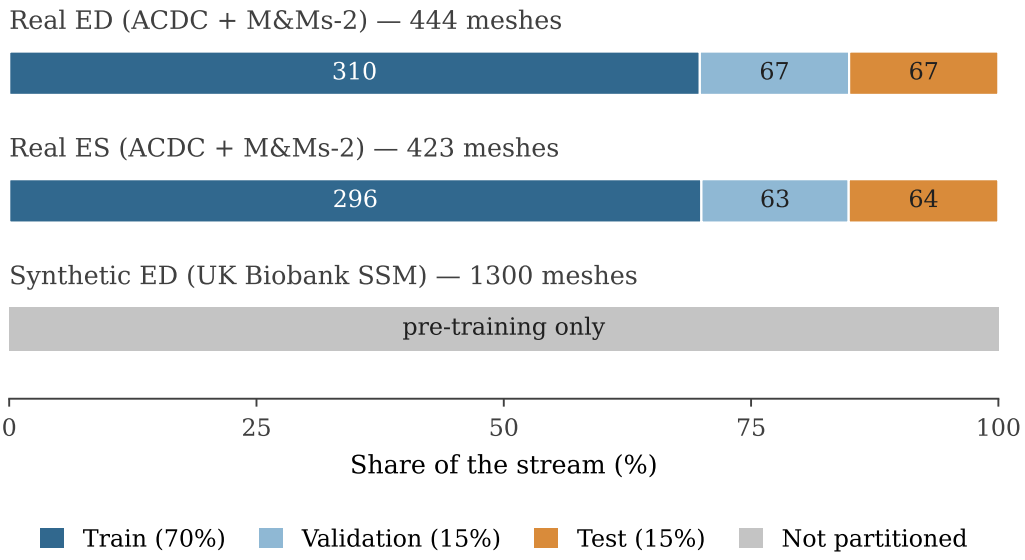



**Figure A.1.** Valid clinical samples divided into the 70 %/15 %/15 % partitions. The 1300 accepted synthetic meshes are reserved for pre-training.

**Table A.2.** Operations used to construct the RBF-derived clinical supervision meshes.

| Operation | Purpose |
|---|---|
| RAS+ canonicalisation | Standardises patient-coordinate orientation while preserving physical spacing |
| Label harmonisation | Maps ACDC and M&Ms-2 labels to common endocardial and epicardial definitions |
| Largest-ring extraction and resampling | Retains 60 ordered points per boundary on each acquired SAX slice |
| In-plane normal estimation | Defines an outward direction at every contour point |
| Off-surface constraints at $\pm 2.5$ mm | Establishes the sign and local scale of each implicit field |
| Thin-plate-spline RBF fit | Interpolates a continuous field with degree 1 and regularisation parameter 0.5 |
| Evaluation on a 1.0 mm grid | Discretises the continuous field for surface extraction |
| Signed-distance regularisation ($\sigma = 1.2$ mm) | Reduces voxel-scale corrugation before surface extraction |
| Connected-region filtering and marching cubes | Removes invalid regions and extracts the zero level set |
| Topology validation and conditional correction | Preserves valid RBF surfaces and corrects only failed shells |
| Geometric plausibility checks | Rejects failed fits and invalid endocardial–epicardial pairs |

**Table A.3.** Purpose of the five training-loss terms.

| Term | Purpose | Effect on reconstruction |
|---|---|---|
| $\mathcal{L}_{\text{surf}}$ | Fits zero-level sets to surface samples | Places endocardial and epicardial surfaces on the target meshes |
| $\mathcal{L}_{\text{SDF}}$ | Fits cached query-point distances | Provides dense volumetric supervision |
| $\mathcal{L}_{\text{eik}}$ | Encourages $\|\nabla f\| = 1$ | Regularises the decoder as a signed-distance field |
| $\mathcal{L}_{\text{off}}$ | Repels off-surface points from zero | Reduces spurious zero crossings away from the myocardium |
| $\mathcal{L}_{\text{WT}}$ | Fits the predicted wall offset to wall targets | Constrains separation of the predicted fields |

**Table A.4.** Main architecture settings of the model.

| Component | Configuration |
|---|---|
| Input features | 3D coordinates, tissue label, and cardiac phase |
| Global encoder type | Shared network with endocardial and epicardial max-pooling |
| Local conditioning | Coarse 3D contour feature volume with trilinear interpolation |
| Latent dimension | 256 |
| Local volume size | $16^3$ with 32 channels |
| Positional encoding | Fourier features with $L = 3$ frequency bands |
| Decoder | 8-layer network, width 512, skip connection at layer 4 |
| Activation | Softplus (smooth) |
| Local injections | Local feature paths into the decoder trunk and offset head |
| Output heads | Endocardial signed distance and positive decoder wall-offset field |
| Offset mapping | Sigmoid component with $\tau_{\text{min}} = 0.05$ and $\delta_{\text{cap}} = 0.45$, plus learned soft-hinge headroom |
| Inference grid | $96^3$ query grid for marching cubes |